CSCM Tool

Test report

Results of the automated verification

Version	4.15.0
Date	2026-10-26
Commit	b6b33f5
Source digest	dec606b768afaa57a8fe4671 (SHA-256 of the source files)

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24. Test Report — CSCM Tool

> **Generated file — do not edit by hand.** Regenerate with `python cscm_tool/scripts/generate_test_report.py` after every software change (Definition of Done, 15-development-standards.md). It runs the complete pytest suite, measures coverage, runs the static quality gates, and appends the run to the history below.

1. Run summary

Verdict	PASS	
Generated (UTC)	2026-10-26 12:29:25	
Branch / commit	dev @ a62d0c6 +uncommitted changes	
Test order	file	
Tests	1007 run — 1007 passed, 0 failed, 0 skipped	
Duration	1209.3 s	
Line coverage (app/)	93% (12577 of 13562 statements)	
Python / platform	3.12.10 / Windows 11	
pytest summary line	`functional: 998 passed, 9 deselected, 5 warnings in 1172.95s (0:19:32)	performance: 9 passed, 1 warning in 51.34s`

2. Results by suite

Suite	Tests	Passed	Failed	Skipped	Time
unit	74	74	0	0	2.8 s
integration	678	678	0	0	909.5 s
system	246	246	0	0	245.9 s
performance	9	9	0	0	51.1 s

3. Static quality and security gates

Gate	Result	Detail
black --check	PASS	259 files would be left unchanged.
ruff check	PASS	All checks passed!
mypy	PASS	Success: no issues found in 156 source files
bandit -r app	PASS	[tester] WARNING nosec encountered (B324), but no failed test on line 74
pip-audit	PASS	runtime packages with advisories: none; dev-only advisories (accepted): black, pytest, starlette

4. Measured performance (NFR-001, NFR-004, TC-024, TC-025)

Test	Measured
test_search_and_read_latency_at_20000_revisions	read_p95_ms = 3.8, revisions = 20000, search_median_ms = 53.5, search_p95_ms = 67.0, speed_factor = 1.0
test_export_of_5000_records_within_target[csv-5]	export_seconds = 0.7, limit_seconds = 5, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[json-5]	export_seconds = 0.93, limit_seconds = 5, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[xlsx-15]	export_seconds = 2.24, limit_seconds = 15, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[pdf-15]	export_seconds = 6.02, limit_seconds = 15, records = 5000, speed_factor = 1.0
test_concurrent_identifier_allocation_in_one_group	allocation_p95_ms = 80.3, speed_factor = 1.0, submissions = 64
test_fifty_concurrent_users_are_served_without_errors	p95_ms_50_users = 74.2, speed_factor = 1.0
test_export_issues_a_constant_number_of_queries	statements_for_two_500_row_exports = 27

Targets: search p95 <= 500 ms and single read p95 <= 200 ms at 20,000 revisions; CSV/JSON export of 5,000 records <= 5 s, XLSX/PDF <= 15 s; 50 concurrent users keep the 500 ms search target; identifier allocation under contention correct and gap-free.

5. Coverage by module

Module	Statements	Missed	Coverage
app/__init__.py	82	0	100%
app/api/__init__.py	0	0	100%
app/api/v1/__init__.py	3	0	100%
app/api/v1/_errors.py	20	2	90%
app/api/v1/audit_routes.py	12	0	100%
app/api/v1/auth_routes.py	60	4	93%
app/api/v1/bulk_routes.py	92	0	100%
app/api/v1/change_request_routes.py	136	11	92%
app/api/v1/domain_signoff_routes.py	60	3	95%
app/api/v1/export_routes.py	77	3	96%
app/api/v1/governance_routes.py	244	3	99%
app/api/v1/health_routes.py	30	2	93%
app/api/v1/import_routes.py	93	11	88%
app/api/v1/lookup_routes.py	70	1	99%
app/api/v1/operations_routes.py	165	0	100%
app/api/v1/package_routes.py	71	8	89%
app/api/v1/publication_routes.py	196	7	96%
app/api/v1/release_routes.py	79	8	90%
app/api/v1/revision_routes.py	26	3	88%
app/api/v1/serializers.py	58	0	100%
app/api/v1/st_import_routes.py	78	4	95%
app/api/v1/status_code_routes.py	211	18	91%
app/api/v1/user_routes.py	96	7	93%
app/audit/__init__.py	0	0	100%
app/audit_chain.py	134	7	95%
app/audit_hash.py	10	0	100%
app/auth/__init__.py	0	0	100%
app/auth/decorators.py	122	3	98%
app/auth/password.py	30	0	100%
app/auth/providers.py	32	1	97%
app/auth/rate_limit.py	34	0	100%
app/auth/roles.py	27	0	100%
app/auth/windows.py	43	2	95%
app/cli.py	84	5	94%
app/config.py	59	0	100%
app/db_modes.py	15	0	100%
app/domain/__init__.py	0	0	100%
app/domain/authz.py	5	0	100%
app/domain/errors.py	29	0	100%
app/domain/lifecycle.py	19	0	100%
app/errors.py	24	3	88%
app/export/__init__.py	0	0	100%
app/export/csv_exporter.py	17	0	100%
app/export/full_database_exporter.py	22	1	95%
app/export/json_exporter.py	3	0	100%
app/export/offline_bundle.py	13	0	100%
app/export/pdf_exporter.py	214	3	99%
app/export/xlsx_exporter.py	20	0	100%
app/extensions.py	9	0	100%

Module	Statements	Missed	Coverage
app/help_content.py	6	0	100%
app/i18n.py	100	9	91%
app/i18n_ar.py	1	0	100%
app/i18n_de.py	3	0	100%
app/i18n_eu.py	4	0	100%
app/i18n_extra.py	10	0	100%
app/i18n_generated.py	13	0	100%
app/i18n_in.py	4	0	100%
app/i18n_messages.py	26	0	100%
app/importers/__init__.py	0	0	100%
app/importers/stcd.py	245	8	97%
app/logging_config.py	45	0	100%
app/models/__init__.py	11	0	100%
app/models/audit.py	16	0	100%
app/models/change_request.py	136	0	100%
app/models/export_job.py	38	0	100%
app/models/import_job.py	30	0	100%
app/models/lookup.py	87	0	100%
app/models/operations.py	121	0	100%
app/models/release.py	30	0	100%
app/models/status_code.py	145	0	100%
app/models/terminology.py	25	0	100%
app/models/translation.py	42	0	100%
app/models/user.py	33	0	100%
app/observability.py	102	11	89%
app/repositories/__init__.py	0	0	100%
app/repositories/audit_repository.py	22	1	95%
app/repositories/base.py	20	1	95%
app/repositories/change_request_repository.py	26	0	100%
app/repositories/fts.py	35	2	94%
app/repositories/release_repository.py	4	0	100%
app/repositories/status_code_repository.py	82	3	96%
app/repositories/user_repository.py	33	1	97%
app/schema_sql.py	45	1	98%
app/security_headers.py	15	0	100%
app/services/__init__.py	0	0	100%
app/services/attachment_service.py	106	11	90%
app/services/audience_service.py	31	1	97%
app/services/audit_service.py	23	0	100%
app/services/auth_service.py	108	9	92%
app/services/backup_service.py	131	7	95%
app/services/baseline_service.py	68	0	100%
app/services/bulk_service.py	354	36	90%
app/services/change_request_service.py	264	13	95%
app/services/change_set_service.py	135	15	89%
app/services/connection_service.py	131	5	96%
app/services/delegation_service.py	86	6	93%
app/services/domain_signoff_service.py	91	5	95%
app/services/duplicate_service.py	41	1	98%
app/services/export_profile_service.py	97	19	80%
app/services/export_service.py	269	8	97%

Module	Statements	Missed	Coverage
app/services/field_review_service.py	126	12	90%
app/services/health_report_service.py	81	4	95%
app/services/hub_consolidation_service.py	106	8	92%
app/services/import_service.py	483	43	91%
app/services/integration_service.py	192	7	96%
app/services/kpi_service.py	72	7	90%
app/services/legacy_mapping_service.py	70	6	91%
app/services/lookup_admin_service.py	96	5	95%
app/services/lookup_service.py	33	4	88%
app/services/mail_service.py	30	3	90%
app/services/manifest_service.py	46	1	98%
app/services/notification_service.py	98	1	99%
app/services/numbering_service.py	78	3	96%
app/services/occurrence_service.py	88	7	92%
app/services/package_service.py	199	27	86%
app/services/reference_service.py	72	11	85%
app/services/release_notes_service.py	115	9	92%
app/services/release_service.py	136	6	96%
app/services/revision_diff_service.py	35	0	100%
app/services/saved_search_service.py	61	0	100%
app/services/settings_service.py	74	6	92%
app/services/sla_service.py	57	0	100%
app/services/software_version_service.py	48	1	98%
app/services/st_import_service.py	546	78	86%
app/services/status_code_service.py	456	19	96%
app/services/terminology_service.py	293	21	93%
app/services/translation_impact_service.py	127	8	94%
app/services/translation_review_service.py	362	33	91%
app/services/translation_service.py	123	10	92%
app/services/user_service.py	162	3	98%
app/services/validation.py	77	9	88%
app/status_code_fields.py	164	6	96%
app/terminology_seed.py	14	1	93%
app/terminology_seed_process.py	2	0	100%
app/terminology_seed_product.py	1	0	100%
app/translation_catalogue.py	123	0	100%
app/translation_precheck.py	141	21	85%
app/ui/__init__.py	73	8	89%
app/ui/admin_views.py	170	21	88%
app/ui/auth_views.py	95	26	73%
app/ui/bulk_views.py	114	17	85%
app/ui/change_request_views.py	192	22	89%
app/ui/dashboard_views.py	34	0	100%
app/ui/export_views.py	23	1	96%
app/ui/governance_views.py	255	40	84%
app/ui/import_views.py	133	15	89%
app/ui/insight_views.py	118	17	86%
app/ui/nav.py	69	2	97%
app/ui/publication_views.py	198	50	75%
app/ui/release_views.py	95	29	69%
app/ui/status_code_views.py	330	55	83%

Module	Statements	Missed	Coverage
app/ui/terminology_views.py	114	5	96%
app/ui/translation_views.py	138	4	97%
app/utils.py	9	0	100%
app/version.py	1	0	100%

6. Traceability evidence check

`docs/20-traceability-matrix.md` cites 585 test names / test files as evidence. **585 resolve** to a test that exists; **0 are stale**.

6a. Requirement coverage

The specification defines **237** requirement ids (FR, NFR, BR in `03-srs.md`, SEC in `11-security-architecture.md`, AUD in `12-audit-compliance.md`). **199** are cited by at least one test (module docstring, test docstring or test name); **37** more are marked covered, with the tests named, in `20-traceability-matrix.md`; **1** are marked there as out of scope or needing a person (`NFR-002`). **0 are not accounted for**.

7. Failures

None.

8. Complete test inventory

Every executed test, grouped by file, with its outcome, duration, and the requirement IDs it cites (module/test docstring or name). Parametrised cases appear individually.

tests/unit/test_lifecycle.py — 52 tests, 52 passed

TC-001 / TC-002 / BR-004: the 7-state lifecycle and its guard.

Result	Test	Purpose	Requirements	Time
OK	test_every_defined_transition_is_accepted[APPROVED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DEPRECATED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DEPRECATED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DRAFT-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[REVIEW-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[REVIEW-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_orm_guard_blocks_an_illegal_status_assignment		BR-004, TC-001, TC-002	0.71 s
OK	test_terminal_and_gated_states_cannot_be_skipped		BR-004, TC-001, TC-002	0.00 s
OK	test_there_are_exactly_seven_states		BR-004, TC-001, TC-002	0.01 s

tests/unit/test_password.py — 5 tests, 5 passed

Result	Test	Purpose	Requirements	Time
OK	test_hash_and_verify_round_trip			0.09 s
OK	test_policy_accepts_reasonable_password			0.00 s
OK	test_policy_rejects_blocklisted_password			0.00 s
OK	test_policy_rejects_short_password			0.00 s
OK	test_verify_rejects_wrong_password			0.08 s

tests/unit/test_rate_limit.py — 6 tests, 6 passed

SEC-004 / I-20: the login rate limiter, backed by the database.

Result	Test	Purpose	Requirements	Time
OK	test_a_refused_attempt_is_not_recorded		NFR-009, SEC-004, TC-050	0.30 s
OK	test_allows_attempts_under_the_limit		NFR-009, SEC-004, TC-050	0.30 s
OK	test_attempts_age_out_of_the_window		NFR-009, SEC-004, TC-050	0.31 s
OK	test_blocks_after_five_username_attempts		NFR-009, SEC-004, TC-050	0.31 s
OK	test_blocks_after_ten_ip_attempts_across_usernames		NFR-009, SEC-004, TC-050	0.31 s
OK	test_username_limit_is_case_insensitive		NFR-009, SEC-004, TC-050	0.31 s

tests/unit/test_validation.py — 11 tests, 11 passed

FR-022, FR-027 field rules from docs/06-data-dictionary.md, v3

Result	Test	Purpose	Requirements	Time
OK	test_description_too_short		FR-022, FR-027	0.00 s
OK	test_number_of_times_repeated_must_be_at_least_one		FR-022, FR-027	0.00 s
OK	test_repeated_over_the_course_format		FR-022, FR-027	0.01 s
OK	test_reset_delay_rejects_missing_unit		FR-022, FR-027	0.00 s
OK	test_set_delay_accepts_leading_star_and_units		FR-022, FR-027	0.00 s
OK	test_set_delay_rejects_missing_unit		FR-022, FR-027	0.00 s
OK	test_status_category_must_be_error_warning_or_info		FR-022, FR-027	0.00 s
OK	test_title_must_not_equal_description		FR-022, FR-027	0.00 s
OK	test_title_too_short		FR-022, FR-027	0.00 s
OK	test_turbine_platforms_required_when_provided_empty		FR-022, FR-027	0.00 s
OK	test_valid_data_has_no_errors		FR-022, FR-027	0.00 s

tests/integration/test_audit_chain_and_ops.py — 27 tests, 27 passed

I-18 tamper-evident audit log, I-09 verified backups, I-21 readiness / metrics / log file.

Result	Test	Purpose	Requirements	Time
OK	test_a_backup_is_written_verified_and_recorded		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.75 s
OK	test_a_configured_key_makes_the_chain_keyed		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.74 s
OK	test_a_corrupt_backup_fails_verification		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.86 s
OK	test_a_forged_row_cannot_be_repaired_without_the_key		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.54 s
OK	test_a_metrics_token_admits_an_address_outside_the_networks		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.45 s
OK	test_a_removed_row_breaks_the_link		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.84 s
OK	test_a_restored_backup_runs_the_application	The restore drill from docs/14 §5 in miniature: back up, start an app on the copy, log in.	AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.93 s
OK	test_a_tampered_audit_log_inside_a_backup_is_flagged		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.76 s
OK	test_an_edited_row_is_found_and_named		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.73 s
OK	test_archive_exports_old_rows_with_a_hash_and_deletes_nothing		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.72 s
OK	test_archive_with_nothing_old_enough_does_nothing		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.85 s

Result	Test	Purpose	Requirements	Time
OK	test_backup_needs_a_file_database	An in-memory database cannot be backed up; built explicitly so the test also holds when	AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.22 s
OK	test_cli_audit_verify_fails_with_a_non_zero_exit_when_broken		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.71 s
OK	test_cli_backup_verify_and_audit_commands		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.74 s
OK	test_every_audit_row_is_chained_to_its_predecessor		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.40 s
OK	test_metrics_are_refused_outside_the_allowed_networks		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.30 s
OK	test_metrics_count_requests_and_hold_no_business_data		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.78 s
OK	test_only_the_newest_generations_are_kept		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.74 s
OK	test_ready_is_503_when_the_chain_is_broken_or_the_schema_outdated		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.82 s
OK	test_ready_needs_no_login_and_reveals_no_business_data		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.35 s
OK	test_ready_reports_ok_and_degraded_by_backup_state		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.12 s
OK	test_requests_are_logged_to_the_configured_file		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.11 s
OK	test_rows_without_hashes_before_the_chain_started_are_tolerated		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.54 s
OK	test_the_append_only_triggers_still_stop_edits_inside_the_application		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.91 s
OK	test_the_hash_is_reproducible_from_the_stored_fields		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.82 s
OK	test_truncating_the_tail_is_caught_by_the_anchor		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.70 s
OK	test_verify_tail_detects_recent_tampering_cheaply		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.72 s

tests/integration/test_audit_coverage.py — 7 tests, 7 passed

FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a: the audit trail

Result	Test	Purpose	Requirements	Time
OK	test_audit_log_entries_cannot_be_updated_or_deleted	FR-072: append-only at the database layer, not just by omission.	FR-072	0.35 s

Result	Test	Purpose	Requirements	Time
OK	test_each_lifecycle_action_is_audited_exactly_once	TC-015: one entry per state-changing action across the whole lifecycle.	TC-015	1.52 s
OK	test_import_lifecycle_is_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.59 s
OK	test_sandbox_delete_keeps_full_snapshot_and_creation_entry	TC-015a / FR-056 / FR-073.	FR-056, FR-073, TC-015a	0.74 s
OK	test_subgroup_creation_is_audited_as_subgroup_create		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.41 s
OK	test_unexpected_import_failure_marks_job_failed_and_is_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.45 s
OK	test_user_and_auth_events_are_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.46 s

tests/integration/test_audit_log.py — 4 tests, 4 passed

FR-070-FR-071: audit log search, Administrator full access,

Result	Test	Purpose	Requirements	Time
OK	test_admin_sees_all_entries		FR-070, FR-071	0.45 s
OK	test_engineer_cannot_view_audit_log		FR-070, FR-071	0.41 s
OK	test_filter_by_entity_type		FR-070, FR-071	0.46 s
OK	test_reviewer_only_sees_own_actions		FR-070, FR-071	0.69 s

tests/integration/test_auth.py — 7 tests, 7 passed

FR-001-FR-004, SEC-001-SEC-004.

Result	Test	Purpose	Requirements	Time
OK	test_deactivated_user_is_immediately_rejected		FR-001, FR-004, SEC-001, SEC-004	0.41 s
OK	test_five_failed_attempts_locks_the_account	The 5th consecutive failure both locks the account (SEC-004) and	SEC-004	0.59 s
OK	test_login_success_returns_user_and_sets_session		FR-001, FR-004, SEC-001, SEC-004	0.39 s
OK	test_logout_invalidates_session		FR-001, FR-004, SEC-001, SEC-004	0.42 s
OK	test_me_requires_login		FR-001, FR-004, SEC-001, SEC-004	0.30 s
OK	test_password_reset_invalidates_existing_session		FR-001, FR-004, SEC-001, SEC-004	0.45 s
OK	test_wrong_password_is_rejected		FR-001, FR-004, SEC-001, SEC-004	0.54 s

tests/integration/test_baseline_and_packages.py — 16 tests, 16 passed

The baseline of a Release and the all-languages / all-access-levels package (docs/03-srs.md

Result	Test	Purpose	Requirements	Time
OK	test_a_hub_controller_group_exists_for_the_controller_filter		BR-007, FR-045, FR-173, FR-174, TC-066	0.48 s
OK	test_a_package_can_be_narrowed_to_a_platform_and_a_controller		BR-007, FR-045, FR-173, FR-174, TC-066	2.33 s
OK	test_a_package_holds_every_language_for_every_access_level		BR-007, FR-045, FR-173, FR-174, TC-066	2.65 s
OK	test_a_package_needs_a_published_release_and_valid_choices		BR-007, FR-045, FR-173, FR-174, TC-066	2.33 s
OK	test_a_release_can_still_be_exported_as_only_its_own_changes		BR-007, FR-045, FR-173, FR-174, TC-066	2.02 s
OK	test_an_unpublished_release_previews_what_its_manual_will_contain		BR-007, FR-045, FR-173, FR-174, TC-066	2.20 s
OK	test_everybody_may_download_what_is_released_and_only_owner_and_admin_what_is_not		BR-007, FR-045, FR-173, FR-174, TC-066	3.45 s
OK	test_the_baseline_narrows_to_a_controller		BR-007, FR-045, FR-173, FR-174, TC-066	1.52 s
OK	test_the_baseline_narrows_to_a_platform		BR-007, FR-045, FR-173, FR-174, TC-066	2.38 s

Result	Test	Purpose	Requirements	Time
OK	test_the_baseline_of_a_release_holds_every_code_valid_at_that_release		BR-007, FR-045, FR-173, FR-174, TC-066	2.10 s
OK	test_the_manifest_vouches_for_every_file_and_reports_the_coverage		BR-007, FR-045, FR-173, FR-174, TC-066	2.49 s
OK	test_the_package_is_audited		BR-007, FR-045, FR-173, FR-174, TC-066	2.10 s
OK	test_the_readiness_check_shows_what_is_still_missing		BR-007, FR-045, FR-173, FR-174, TC-066	2.25 s
OK	test_the_release_page_offers_the_baseline_and_the_package		BR-007, FR-045, FR-173, FR-174, TC-066	2.88 s
OK	test_the_summary_counts_the_baseline_by_controller_and_platform		BR-007, FR-045, FR-173, FR-174, TC-066	2.35 s
OK	test_the_title_page_names_the_scope		BR-007, FR-045, FR-173, FR-174, TC-066	2.62 s

tests/integration/test_bulk_workflow.py — 19 tests, 19 passed

Bulk work on a change set with a sample inspection (docs/03-srs.md FR-180): edit, submit, domain

Result	Test	Purpose	Requirements	Time
OK	test_a_bulk_domain_signoff_needs_a_sample_and_every_code_gets_its_own_signoff		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.15 s
OK	test_a_decision_in_the_workbench_names_its_sample		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.73 s
OK	test_a_defect_in_the_sample_blocks_the_decision_until_it_is_sent_back		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.95 s
OK	test_a_defect_needs_a_note_and_a_verdict_must_be_ok_or_defect		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.59 s
OK	test_a_domain_signoff_is_only_for_a_holder_of_the_domain		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.92 s
OK	test_a_sample_cannot_be_borrowed_reused_or_kept_after_the_batch_changed		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.94 s
OK	test_a_small_batch_needs_no_sample		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.93 s
OK	test_a_whole_domain_signoff_can_be_done_in_the_workbench		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.43 s
OK	test_bulk_edit_refuses_unknown_fields_and_other_peoples_drafts		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.09 s
OK	test_bulk_edit_sets_the_same_value_on_every_draft_with_its_own_audit_entry		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.99 s
OK	test_bulk_submit_sends_the_drafts_that_are_ready_and_reports_the_rest		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.17 s
OK	test_the_bulk_screens_are_translated		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.00 s
OK	test_the_dossier_puts_the_codes_the_samples_and_the_audit_in_one_workbook		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.50 s
OK	test_the_person_who_wrote_a_code_is_not_part_of_the_batch_they_decide		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.83 s
OK	test_the_readiness_names_what_stops_each_draft		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.90 s
OK	test_the_sample_is_spread_over_the_subgroups_and_reproducible		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.00 s
OK	test_the_size_of_the_sample_follows_the_settings		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.66 s
OK	test_the_workbench_shows_the_batch_the_readiness_and_the_rule		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.30 s
OK	test_viewers_see_nothing_of_the_bulk_workbench		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.79 s

tests/integration/test_change_requests.py — 11 tests, 11 passed

FR-030-FR-039, FR-027-FR-029, BR-004/BR-005. Full submit -> dual-review ->

Result	Test	Purpose	Requirements	Time
OK	test_author_cannot_sign_off_own_cr		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.93 s
OK	test_chief_engineer_cannot_be_a_prior_signer		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.21 s
OK	test_domain_signoff_rejects_user_without_that_domain		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.58 s
OK	test_dual_signoff_in_either_order_reaches_pending_approval		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.02 s
OK	test_full_happy_path_to_approved		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.17 s
OK	test_rejection_returns_to_draft_and_clears_signoffs		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.07 s
OK	test_reviewer_rejection_without_comment_is_rejected		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.85 s
OK	test_submit_blocked_until_domain_signoff_complete		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.87 s
OK	test_withdraw_forbidden_for_non_author		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.59 s
OK	test_withdraw_removes_abandoned_new_draft		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.69 s
OK	test_withdraw_revision_type_restores_prior_current_revision	Regression test: withdrawing a REVISION-type CR must restore the	BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.45 s

tests/integration/test_connections.py — 12 tests, 12 passed

How status codes are connected: suppression by a cause, preconditions and escalation, read from the

Result	Test	Purpose	Requirements	Time
OK	test_a_negated_active_flag_is_a_cause_and_a_plain_one_a_precondition		FR-048o, FR-092, FR-179, TC-069	0.00 s
OK	test_a_viewer_does_not_see_the_connections		FR-048o, FR-092, FR-179, TC-069	0.94 s
OK	test_an_imported_program_brings_its_cause_and_its_repeat_code		FR-048o, FR-092, FR-179, TC-069	0.86 s
OK	test_escalation_targets_say_which_are_outside_the_catalogue		FR-048o, FR-092, FR-179, TC-069	0.73 s
OK	test_one_code_shows_what_it_depends_on_and_what_depends_on_it		FR-048o, FR-092, FR-179, TC-069	0.96 s
OK	test_the_api_returns_the_connections_of_a_code_and_of_the_catalogue		FR-048o, FR-092, FR-179, TC-069	0.80 s
OK	test_the_code_page_and_the_overview_show_the_connections		FR-048o, FR-092, FR-179, TC-069	1.28 s
OK	test_the_hierarchy_puts_the_cause_on_top_with_what_it_suppresses_below		FR-048o, FR-092, FR-179, TC-069	0.74 s
OK	test_the_hub_and_main_controller_can_be_looked_at_separately		FR-048o, FR-092, FR-179, TC-069	0.73 s
OK	test_the_menu_offers_the_connections_to_the_working_roles_only		FR-048o, FR-092, FR-179, TC-069	0.92 s
OK	test_the_real_programs_name_their_cause_and_their_repeat_code		FR-048o, FR-092, FR-179, TC-069	0.01 s
OK	test_what_looks_wrong_is_listed		FR-048o, FR-092, FR-179, TC-069	0.70 s

tests/integration/test_content_integrity.py — 16 tests, 16 passed

I-01 optimistic concurrency, I-04 reference warnings and where-used, I-12 duplicate

Result	Test	Purpose	Requirements	Time
OK	test_expected_updated_at_is_enforced_too		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.81 s
OK	test_next_available_reports_how_many_numbers_are_left		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.26 s
OK	test_planned_dates_are_administrator_set_audited_and_warn_at_publish		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.38 s
OK	test_platform_variant_copies_the_code_and_links_the_family		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.62 s
OK	test_reference_to_a_deprecated_code_warns		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	2.10 s

Result	Test	Purpose	Requirements	Time
OK	test_related_codes_are_validated_and_checked		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.82 s
OK	test_remedy_text_length_is_validated		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.69 s
OK	test_save_without_a_version_still_works_for_old_clients		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.83 s
OK	test_similar_endpoint_supports_the_did_you_mean_hint		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.83 s
OK	test_similar_title_in_the_subgroup_is_flagged_not_blocked		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.08 s
OK	test_stale_save_is_refused_with_the_current_values		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.18 s
OK	test_troubleshooting_fields_round_trip_and_reach_the_export		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.56 s
OK	test_ui_stale_edit_keeps_the_users_entries_and_shows_the_conflict		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.10 s
OK	test_unknown_escalation_number_warns_but_does_not_block		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.06 s
OK	test_variant_needs_a_decided_source_and_an_engineer		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	0.83 s
OK	test_where_used_lists_dependants_and_deprecation_needs_acknowledgement		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.62 s

tests/integration/test_csrf.py — 2 tests, 2 passed

docs/11-security-architecture.md §5: CSRF synchronizer-token protection

Result	Test	Purpose	Requirements	Time
OK	test_state_changing_request_with_csrf_token_succeeds			0.41 s
OK	test_state_changing_request_without_csrf_token_is_rejected			0.66 s

tests/integration/test_decision_trail.py — 5 tests, 5 passed

Where review comments live: the change-request page, the status-code

Result	Test	Purpose	Requirements	Time
OK	test_audit_entries_carry_the_comment_text_and_id			1.29 s
OK	test_audit_log_can_be_filtered_by_action_and_actor			1.30 s
OK	test_status_code_page_lists_all_decisions_and_comments			1.40 s
OK	test_viewers_cannot_see_the_review_discussion			1.42 s
OK	test_withdrawing_keeps_the_discussion_in_the_audit_log			1.16 s

tests/integration/test_deprecation.py — 7 tests, 7 passed

FR-050-FR-052, BR-006, docs/05-governance-handbook.md §5, §11.

Result	Test	Purpose	Requirements	Time
OK	test_archive_blocked_before_retention_period_elapses		BR-006, FR-050, FR-052	1.27 s
OK	test_archive_succeeds_after_retention_period		BR-006, FR-050, FR-052	1.43 s
OK	test_deprecation_requires_released_state		BR-006, FR-050, FR-052	0.69 s
OK	test_full_deprecation_cycle		BR-006, FR-050, FR-052	1.72 s
OK	test_non_admin_cannot_archive_or_reinstate		BR-006, FR-050, FR-052	0.43 s
OK	test_reinstate_deprecated_code		BR-006, FR-050, FR-052	1.65 s
OK	test_second_revision_does_not_violate_current_uniqueness	Regression test: create_revision must flip the old current revision's	BR-006, FR-050, FR-052	1.44 s

tests/integration/test_email_and_windows_signin.py — 16 tests, 16 passed

The e-mail address of a user is fixed, and the Windows sign-in that will replace the password in a

Result	Test	Purpose	Requirements	Time
OK	test_a_domain_name_is_matched_through_the_mail_domain_or_the_user_name		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.70 s
OK	test_a_header_from_an_address_that_is_not_a_trusted_proxy_is_not_believed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.70 s
OK	test_an_address_is_stored_trimmed_and_lower_case_and_must_be_valid_and_new		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.85 s
OK	test_no_identity_an_unknown_one_and_a_deactivated_account_are_all_refused		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.79 s
OK	test_production_never_starts_with_the_simulation_or_an_unknown_mode		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.10 s
OK	test_the_address_of_a_user_cannot_be_changed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.76 s
OK	test_the_database_refuses_a_changed_address_too		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.70 s
OK	test_the_development_defaults_are_the_simulation_next_to_the_password		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.18 s
OK	test_the_gateways_identity_signs_the_matching_account_in_and_is_audited		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.07 s
OK	test_the_screens_show_the_address_as_fixed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.91 s
OK	test_the_shipped_example_configuration_documents_the_settings		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.01 s
OK	test_the_simulation_page_stands_in_for_the_gateway_and_signs_in_by_address		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.05 s
OK	test_the_simulation_refuses_an_unknown_identity_and_is_off_by_default		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.73 s
OK	test_with_both_modes_the_login_page_offers_the_button_and_the_form		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.34 s
OK	test_with_passwords_only_the_windows_sign_in_is_not_there		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.81 s
OK	test_with_windows_only_passwords_are_refused_and_the_login_page_says_so		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.79 s

tests/integration/test_export_language.py — 4 tests, 4 passed

The words around an export follow the language of the export (docs/03-srs.md FR-171): the title

Result	Test	Purpose	Requirements	Time
OK	test_a_pdf_in_french_prints_its_own_words_in_french		FR-045, FR-110, FR-165, FR-171, TC-065	0.06 s
OK	test_a_placeholder_survives_in_every_language		FR-045, FR-110, FR-165, FR-171, TC-065	0.00 s
OK	test_english_and_no_language_leave_the_text_as_it_is		FR-045, FR-110, FR-165, FR-171, TC-065	0.05 s
OK	test_every_word_around_an_export_exists_in_every_language		FR-045, FR-110, FR-165, FR-171, TC-065	0.00 s

tests/integration/test_exports_and_diff.py — 15 tests, 15 passed

FR-032, FR-045, FR-046, FR-054, FR-064, BR-008, SEC-023, TC-023, TC-035:

Result	Test	Purpose	Requirements	Time
OK	test_diff_card_renders_on_change_request_page		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.44 s
OK	test_diff_hidden_from_viewers_for_open_revisions_and_unknown_revisions		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.62 s

Result	Test	Purpose	Requirements	Time
OK	test_export_file_names_carry_a_timestamp	Every export file name ends with the generation time, YYYY-MM-DD_HH-MM (UTC).	BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.69 s
OK	test_exports_neutralise_formula_injection		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.04 s
OK	test_json_export_is_flat_rows		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.56 s
OK	test_new_code_diff_is_marked_new		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	0.82 s
OK	test_release_can_be_deleted_before_publish_only		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.80 s
OK	test_released_export_contains_only_released_non_sandbox_codes		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.68 s
OK	test_released_export_filters_and_formats		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.92 s
OK	test_revision_diff_endpoint_and_cr_baseline_diff		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.63 s
OK	test_working_set_filters_by_lifecycle_state		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	0.91 s
OK	test_working_set_is_labelled_and_excludes_sandbox		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.44 s
OK	test_working_set_not_available_to_viewers_or_anonymous		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	0.94 s
OK	test_working_set_pdf_and_xlsx_render		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.24 s
OK	test_xlsx_export_has_same_rows_as_csv	TC-035.	TC-035	1.63 s

tests/integration/test_generated_fields.py — 20 tests, 20 passed

The fields of the generated controller programs (schema v4.1, docs/06-data-dictionary.md sections

Result	Test	Purpose	Requirements	Time
OK	test_a_field_of_the_other_controller_is_refused_and_the_rest_is_stored		FR-025, FR-032, FR-048o, FR-168, TC-063	1.05 s
OK	test_a_hub_program_import_keeps_its_hub_fields_and_nothing_of_the_main_ones		FR-025, FR-032, FR-048o, FR-168, TC-063	0.79 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1000.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.16 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1001.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.81 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1002.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.79 s
OK	test_a_named_audience_value_brings_its_default_levels_and_explicit_levels_win		FR-025, FR-032, FR-048o, FR-168, TC-063	0.80 s
OK	test_a_new_revision_carries_over_what_the_caller_does_not_send_and_the_diff_shows_changes		FR-025, FR-032, FR-048o, FR-168, TC-063	1.26 s
OK	test_a_rejected_form_keeps_what_was_typed		FR-025, FR-032, FR-048o, FR-168, TC-063	0.85 s
OK	test_the_csv_import_carries_the_new_columns_and_an_update_without_them_keeps_the_values		FR-025, FR-032, FR-048o, FR-168, TC-063	0.85 s
OK	test_the_form_stores_the_fields_and_the_pages_show_them		FR-025, FR-032, FR-048o, FR-168, TC-063	1.07 s
OK	test_the_levels_and_programs_are_internal_and_never_leave_in_an_audience_export		FR-025, FR-032, FR-048o, FR-168, TC-063	1.72 s
OK	test_the_two_zero_load_attributes_are_not_mixed_up		FR-025, FR-032, FR-048o, FR-168, TC-063	0.78 s
OK	test_values_out_of_range_are_refused[extra0]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.72 s

Result	Test	Purpose	Requirements	Time
OK	test_values_out_of_range_are_refused[extra1]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.72 s
OK	test_values_out_of_range_are_refused[extra2]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.71 s
OK	test_values_out_of_range_are_refused[extra3]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.69 s
OK	test_values_out_of_range_are_refused[extra4]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.72 s
OK	test_values_out_of_range_are_refused[extra5]		FR-025, FR-032, FR-048o, FR-168, TC-063	0.72 s
OK	test_values_out_of_range_are_refused[extra6]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.00 s
OK	test_withdrawing_and_deleting_a_code_removes_its_levels		FR-025, FR-032, FR-048o, FR-168, TC-063	0.83 s

tests/integration/test_governance_extras.py — 24 tests, 24 passed

Remaining spec checks: SEC-010 service-layer authorization, FR-010/011/011a/014

Result	Test	Purpose	Requirements	Time
OK	test_a_user_holds_exactly_one_role		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.52 s
OK	test_account_lock_and_password_events_are_audited		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.92 s
OK	test_auth_provider_seam_is_substitutable	Callers depend on AuthenticatedUser only: a different provider returning	FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.81 s
OK	test_comment_leaves_state_and_signoffs_untouched		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.11 s
OK	test_import_preserves_legacy_reference_number		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.84 s
OK	test_next_available_preview_does_not_reserve_a_number		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.08 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ADMINISTRATOR]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.66 s
OK	test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.30 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER2]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER3]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.30 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER4]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER5]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.32 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER6]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.32 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER7]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.32 s
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s

Result	Test	Purpose	Requirements	Time
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER2]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.31 s
OK	test_services_enforce_roles_independently_of_route_decorators[-VIEWER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.67 s
OK	test_services_enforce_roles_independently_of_route_decorators[-VIEWER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.30 s
OK	test_user_administration_events_are_audited		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.70 s
OK	test_whole_lifecycle_works_with_all_outbound_network_blocked	The application must keep operating when it can reach nothing beyond	FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	2.09 s

tests/integration/test_import.py — 7 tests, 7 passed

FR-048a-d, EPIC-14: CSV/XLSX library import, still fully governed

Result	Test	Purpose	Requirements	Time
OK	test_ambiguous_group_code_requires_range_hint	WTUR is reused for a second, numerically distinct Hub Controller	FR-048a	0.78 s
OK	test_imported_row_becomes_import_type_draft_change_request		FR-048a	0.47 s
OK	test_invalid_row_is_reported_and_creates_nothing		FR-048a	0.50 s
OK	test_non_admin_engineer_can_upload_import		FR-048a	0.46 s
OK	test_unknown_functional_system_group_code_reported_as_error		FR-048a	0.45 s
OK	test_unsupported_file_extension_rejected		FR-048a	0.44 s
OK	test_viewer_cannot_upload_import		FR-048a	0.42 s

tests/integration/test_import_toolkit.py — 20 tests, 20 passed

The migration toolkit around the library import (docs/21-improvement-ideas.md): I-05 template, dry

Result	Test	Purpose	Requirements	Time
OK	test_a_big_file_is_processed_in_the_background_with_progress		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.60 s
OK	test_a_legacy_number_already_imported_is_refused_in_create_mode		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.55 s
OK	test_dry_run_reports_every_problem_and_creates_nothing		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.44 s
OK	test_dry_run_warns_about_a_possible_duplicate_without_blocking		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.48 s
OK	test_error_report_has_row_column_value_and_a_fix_hint_as_csv_and_xlsx		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.49 s
OK	test_files_above_the_threshold_go_to_the_background_automatically		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.46 s
OK	test_formula_like_values_in_the_error_report_are_neutralised		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.48 s
OK	test_import_ui_pages_render_and_the_validate_button_creates_nothing		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.92 s
OK	test_missing_required_columns_are_named_up_front		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.74 s
OK	test_new_content_columns_are_imported		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.76 s
OK	test_reconciliation_counts_states_and_names_the_missing_numbers		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.52 s
OK	test_similar_titles_are_reported_as_warnings		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.47 s
OK	test_template_csv_and_xlsx_carry_the_exact_headers_and_lookup_codes		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.53 s

Result	Test	Purpose	Requirements	Time
OK	test_the_downloaded_template_imports_cleanly_as_is		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.47 s
OK	test_the_same_legacy_number_in_another_group_is_a_different_code		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.49 s
OK	test_update_mode_errors_are_specific		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.11 s
OK	test_update_mode_opens_a_new_revision_of_a_released_code		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.52 s
OK	test_update_mode_revises_the_open_draft_in_place		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.52 s
OK	test_update_mode_skips_unchanged_rows_and_creates_missing_ones		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.52 s
OK	test_update_of_someone_elses_draft_is_reported_on_the_row		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.94 s

tests/integration/test_integrations_and_reports.py — 32 tests, 32 passed

I-29 API tokens, change feed and webhooks; I-26 saved searches; I-28 fleet occurrences;

Result	Test	Purpose	Requirements	Time
OK	test_a_disabled_webhook_is_not_called		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.70 s
OK	test_a_failing_webhook_is_recorded_and_never_blocks_the_publish		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.87 s
OK	test_a_hub_code_is_carried_over_as_a_main_controller_draft		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.59 s
OK	test_a_publish_calls_the_webhook_with_a_verifiable_signature		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.86 s
OK	test_a_token_is_shown_once_and_stored_only_as_a_hash		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.72 s
OK	test_a_token_only_works_for_its_scope		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.70 s
OK	test_a_token_reads_released_data_only_and_never_writes		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.42 s
OK	test_an_unreachable_webhook_target_does_not_block_the_publish		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	4.26 s
OK	test_bulk_deprecation_is_for_administrators_and_hub_groups		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.90 s
OK	test_carry_over_is_refused_for_the_wrong_kind_of_code_or_target		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.84 s
OK	test_health_report_covers_capacity_missing_data_and_inconsistencies		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.68 s
OK	test_health_report_finds_an_info_code_with_an_alarm		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.22 s
OK	test_health_report_flags_a_nearly_full_band		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.38 s
OK	test_hub_consolidation_ui_page_renders		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.45 s
OK	test_kpis_have_the_documented_shape		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.42 s
OK	test_occurrence_import_validation_and_permissions		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.82 s
OK	test_occurrence_report_names_codes_that_never_fire_and_those_that_fire_most		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.00 s
OK	test_redirects_are_not_followed		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.01 s
OK	test_reports_are_limited_to_the_right_roles		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.06 s
OK	test_revoked_expired_unknown_and_orphaned_tokens_are_refused		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.81 s

Result	Test	Purpose	Requirements	Time
OK	test_sandbox_activity_never_appears_in_the_feed		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.74 s
OK	test_saved_search_ui_save_and_delete		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.82 s
OK	test_saved_searches_are_private_to_their_owner		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.81 s
OK	test_saved_searches_round_trip_through_the_url		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.88 s
OK	test_saving_under_an_existing_name_replaces_it_and_validates_the_name		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.01 s
OK	test_the_capacity_table_colours_main_blue_and_hub_light_blue		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.79 s
OK	test_the_feed_lists_releases_in_order_with_a_cursor_and_an_etag		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.96 s
OK	test_the_feed_needs_authentication		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.29 s
OK	test_the_whole_consolidation_loop_ends_with_a_deprecation_request		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.15 s
OK	test_token_events_are_audited_without_the_secret		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.75 s
OK	test_token_management_is_for_administrators_and_validated		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.75 s
OK	test_webhook_targets_must_be_internal_and_inputs_valid		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.82 s

tests/integration/test_invariants.py — 15 tests, 15 passed

Business-rule and data-integrity checks: BR-001..BR-003, BR-009, FR-023a,

Result	Test	Purpose	Requirements	Time
OK	test_draft_can_be_saved_repeatedly_without_workflow_transitions		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	0.85 s
OK	test_duplicate_identifier_rejected_by_database		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	0.91 s
OK	test_each_subgroup_sub_band_numbers_independently		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	0.31 s
OK	test_exactly_one_current_revision_per_code	BR-002 / TC-011.	BR-002, TC-011	1.87 s
OK	test_filters_sort_and_pagination		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.14 s
OK	test_full_text_search_matches_identifier_title_and_description		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.43 s
OK	test_identifier_formats		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.37 s
OK	test_identifier_never_changes_after_allocation	BR-001: a rejected-and-resubmitted CR keeps its identifier.	BR-001	1.26 s
OK	test_in_flight_revision_cannot_be_edited_in_place	FR-025: Review and later states are locked.	FR-025	1.00 s
OK	test_new_revision_of_a_released_code_keeps_its_identifier	BR-001 / FR-021: the identifier is allocated once, not again per revision.	BR-001, FR-021	1.72 s
OK	test_only_one_open_revision_per_code	BR-003 / TC-012: a second revision cannot be opened while one is open.	BR-003, TC-012	1.52 s
OK	test_released_code_stays_visible_to_viewers_while_a_new_revision_is_open	BR-009: opening revision 2 (a Draft) must not make the Released revision 1	BR-009	2.04 s
OK	test_revision_history_visible_to_viewer_only_once_decided	FR-064 / BR-009.	BR-009, FR-064	1.65 s

Result	Test	Purpose	Requirements	Time
OK	test_sequential_submits_in_two_subgroups_do_not_interfere		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.56 s
OK	test_submit_requires_a_turbine_platform		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	0.77 s

tests/integration/test_lookup_admin.py — 9 tests, 9 passed

US-091: Administrator manages controlled vocabularies. docs/05-governance-

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_add_turbine_platform		US-091	0.43 s
OK	test_admin_can_create_new_functional_system_group_in_free_range	FR-048g: into any currently free range (10000-10999, 14000-14999,	FR-048g	0.42 s
OK	test_admin_can_deactivate_and_relabel_a_value		US-091	0.44 s
OK	test_create_functional_subgroup_with_range		US-091	0.85 s
OK	test_duplicate_code_rejected		US-091	0.44 s
OK	test_functional_subgroup_overlap_rejected	FR-048f: rejected with 409, naming the conflicting subgroup.	FR-048f	0.43 s
OK	test_functional_system_group_not_creatable_via_generic_endpoint	v3 (EPIC-16): creation moved to a dedicated, non-overlap-validated	US-091	0.43 s
OK	test_new_functional_system_group_range_overlap_rejected	FR-048h: rejected with 409, naming the conflicting group.	FR-048h	0.42 s
OK	test_non_admin_cannot_create_lookup_value		US-091	0.44 s

tests/integration/test_menu.py — 22 tests, 22 passed

The menu on the left (docs/10-ui-ux-specification.md section 3, docs/03-srs.md FR-169): what each role

Result	Test	Purpose	Requirements	Time
OK	test_a_review_removes_the_mark_and_a_changed_string_brings_it_back		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.35 s
OK	test_a_viewer_cannot_choose_or_keep_a_language_nobody_has_reviewed		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.00 s
OK	test_a_viewer_sees_only_the_reviewed_languages_and_nothing_marked		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.16 s
OK	test_a_viewer_sees_the_catalogue_and_nothing_to_work_with		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.80 s
OK	test_a_viewer_who_reviews_a_language_sees_that_one		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.81 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[admin]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.82 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[chief_engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.84 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.82 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[reviewer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.83 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[stcd_owner]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.16 s
OK	test_groups_can_be_closed_and_a_narrow_window_has_a_menu_button		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.86 s
OK	test_no_link_appears_twice_and_the_groups_are_the_planned_ones		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.83 s
OK	test_only_the_link_of_the_current_page_is_marked		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.17 s
OK	test_the_login_page_offers_only_the_reviewed_languages		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.27 s
OK	test_the_marker_can_be_switched_off_in_the_settings		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.01 s

Result	Test	Purpose	Requirements	Time
OK	test_the_menu_is_translated_and_a_group_title_too		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.82 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[admin]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.83 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[chief_engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.82 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.15 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[reviewer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.83 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[stcd_owner]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.81 s
OK	test_what_waits_for_the_user_is_counted_in_the_menu		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.19 s

tests/integration/test_numbering.py — 4 tests, 4 passed

docs/07-database-design.md §7a: atomic Functional Group/Subgroup allocation.

Result	Test	Purpose	Requirements	Time
OK	test_allocation_increments_and_persists			0.37 s
OK	test_capacity_exhaustion_raises			0.38 s
OK	test_first_allocation_in_a_group_starts_at_range_start			0.32 s
OK	test_sandbox_allocation_never_collides_with_real_range			0.29 s

tests/integration/test_pdf_scripts.py — 20 tests, 20 passed

The PDF export prints every script of the languages the tool offers (docs/03-srs.md FR-165,

Result	Test	Purpose	Requirements	Time
OK	test_a_large_export_stays_fast_for_latin_text		FR-045, FR-157, FR-165, NFR-008, TC-060	6.78 s
OK	test_a_plain_latin_export_loads_only_noto_sans		FR-045, FR-157, FR-165, NFR-008, TC-060	0.03 s
OK	test_a_script_gets_its_own_font[arabic-\u0631\u0645\u0632 \u0627\u0644\u062d\u0627\u0644\u0629 \u0648\u062d\u062f\u0629 \u0627\u0644\u062a\u062d\u0643\u0645 StCd-01000]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[bengali-\u09b8\u09cd\u09a5\u09bf\u09a4\u09bf \u0995\u09cb\u09a1 \u09a8\u09bf\u09af\u09bc\u09a8\u09cd\u09a4\u09cd\u09b0\u0995 \u09aa\u09b0\u09c0\u0995\u09cd\u09b7\u09be]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[devanagari-\u0938\u094d\u0925\u093f\u0924\u093f \u0915\u094b\u0921 \u0928\u093f\u092f\u0902\u0924\u094d\u0930\u0915 \u0915\u0940 \u091c\u093e\u0901\u091a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[georgian-\u10e1\u10e2\u10d0\u10e2\u10e3\u10e1\u10d8\u10e1 \u10d9\u10dd\u10d3\u10d8 \u10d9\u10dd\u10dc\u10e2\u10e0\u10dd\u10da\u10d4\u10e0\u10d8 \u10e1 \u10e8\u10d4\u10db\u10dd\u10ec\u10db\u10d4\u10d1\u10d0]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[gujarati-\u0ab8\u0acd\u0aa5\u0abf\u0aa4\u0abf \u0a95\u0acb\u0aa1 \u0aa8\u0abf\u0aaf\u0a82\u0aa4\u0acd\u0ab0\u0a95 \u0aa4\u0aaa\u0abe\u0ab8]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[gurmukhi-\u0a38\u0a25\u0a3f\u0a24 \u0a40 \u0a15\u0a4b\u0a21 \u0a15\u0a70\u0a1f\u0a30\u0a4b\u0a32\u0a30 \u0a1c\u0a3e\u0a02\u0a1a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[kannada-\u0cb8\u0ccd\u0ca5\u0cbf\u0ca4\u0cbf \u0c95\u0ccb\u0ca1\u0ccd \u0ca8\u0cbf\u0caf\u0c82\u0ca4\u0ccd\u0cb0\u0c95 \u0caa\u0cb0\u0cbf\u0cb6\u0cc0\u0cb2\u0ca8\u0cc6]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_a_script_gets_its_own_font[malayalam-\u0d38\u0d4d\u0d25\u0d3f\u0d24\u0d3f \u0d15\u0d4b\u0d21\u0d4d \u0d28\u0d3f\u0d2f\u0d28\u0d4d\u0d24\u0d4d\u0d30\u0d15\u0d02 \u0d2a\u0d30\u0d3f\u0d36\u0d4b\u0d27\u0d28]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[odia-\u0b38\u0b4d\u0b25\u0b3f\u0b24\u0b3f \u0b15\u0b4b\u0b21\u0b4d \u0b28\u0b3f\u0b5f\u0b28\u0b4d\u0b24\u0b4d\u0b30\u0b15 \u0b2f\u0b3e\u0b1e\u0b4d\u0b1a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[tamil-\u0ba8\u0bbf\u0bb2\u0bc8 \u0b95\u0bc1\u0bb1\u0bbf\u0baf\u0bc0\u0b9f\u0bc1 \u0b95\u0b9f\u0bcd\u0b9f\u0bc1\u0baa\u0bcd\u0baa\u0b9f\u0bc1\ u0ba4\u0bcd\u0ba4\u0bbf \u0b9a\u0bcb\u0ba4\u0ba9\u0bc8]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[telugu-\u0c38\u0c4d\u0c25\u0c3f\u0c24\u0c3f \u0c15\u0c4b\u0c21\u0c4d \u0c28\u0c3f\u0c2f\u0c02\u0c24\u0c4d\u0c30\u0c3f\u0c15 \u0c24\u0c28\u0c3f\u0c16\u0c40]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[urdu-\u0627\u0633\u0679\u06cc\u0679\u0633 \u06a9\u0648\u0688 \u06a9\u0646\u0679\u0631\u0648\u0644\u0631 \u06a9\u06cc \u062c\u0627\u0646\u0686]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_every_character_of_every_translation_is_drawn_by_a_bundled_font		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_every_script_builds_a_pdf_with_its_font		FR-045, FR-157, FR-165, NFR-008, TC-060	0.35 s
OK	test_latin_text_needs_no_script_font_and_arabic_is_told_from_urdu		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_shaping_is_switched_on_per_cell_only_where_a_script_needs_it		FR-045, FR-157, FR-165, NFR-008, TC-060	0.05 s
OK	test_the_fonts_and_their_licence_are_bundled_and_shaping_is_available		FR-045, FR-157, FR-165, NFR-008, TC-060	0.02 s
OK	test_the_title_page_and_notice_may_carry_any_script		FR-045, FR-157, FR-165, NFR-008, TC-060	0.11 s

tests/integration/test_prg_idx_and_user_groups.py — 9 tests, 9 passed

_Schema v4.1 (migration c9e2a4b6d815): PrgIdx_GUi on the status code, the controller-specific_

Result	Test	Purpose	Requirements	Time
OK	test_default_levels_are_seeded_only_where_the_programs_agree			0.30 s
OK	test_leaving_draft_without_prg_idx_is_blocked_only_once_the_switch_is_on			0.31 s
OK	test_new_revision_columns_start_at_zero_or_not_captured			0.30 s
OK	test_prg_idx_is_an_optional_uint_that_is_unique_among_real_codes			0.32 s
OK	test_prg_idx_is_frozen_once_a_revision_was_released			0.63 s
OK	test_the_locked_revision_trigger_covers_the_new_content_columns			0.31 s
OK	test_the_migration_is_reversible_and_re_applies_cleanly			0.73 s
OK	test_the_sixteen_user_group_slots_map_slots_one_to_eight_to_the_audiences			0.30 s
OK	test_user_group_levels_are_bytes_and_unique_per_revision_and_slot			0.30 s

tests/integration/test_publication.py — 25 tests, 25 passed

Publishing the catalogue (docs/21-improvement-ideas.md): I-02 profiles and wrapped PDF, I-03

Result	Test	Purpose	Requirements	Time
OK	test_administrator_manages_profiles_as_data		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.85 s
OK	test_applicability_matrix_and_changes_between_versions		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.88 s

Result	Test	Purpose	Requirements	Time
OK	test_audience_export_only_contains_codes_the_audience_may_see		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.56 s
OK	test_bundle_contains_the_file_and_a_manifest_that_matches		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.73 s
OK	test_covert_visibility_is_data_an_administrator_can_flip		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.42 s
OK	test_customer_profile_applies_its_default_audience_and_working_set_refuses_audiences		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.97 s
OK	test_draft_source_text_cannot_be_translated_yet		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.77 s
OK	test_export_history_lists_hashes_per_release		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.82 s
OK	test_export_profile_fields_match_the_export_row		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.27 s
OK	test_offline_bundle_is_one_self_contained_page_with_the_data		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.41 s
OK	test_offline_bundle_respects_audiences_and_needs_a_published_release		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.92 s
OK	test_pdf_cells_are_wrapped_not_truncated		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.56 s
OK	test_pdf_column_widths_fit_the_page		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.63 s
OK	test_pdf_survives_characters_no_font_has		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.35 s
OK	test_publication_ui_pages_render		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.82 s
OK	test_release_notes_list_added_changed_and_deprecated		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	4.08 s
OK	test_release_targets_limit_the_candidates_and_freeze_at_publish		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.85 s
OK	test_signed_manifest_detects_edits_to_the_manifest_itself		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.40 s
OK	test_three_profiles_are_seeded_and_shape_the_columns		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.86 s
OK	test_translation_validation_worklist_and_carry_over		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.40 s
OK	test_translation_workflow_with_light_review_and_language_exports		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.32 s
OK	test_unpublished_release_notes_are_an_administrator_preview		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.29 s
OK	test_verify_export_cli_works_without_the_database		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.41 s

Result	Test	Purpose	Requirements	Time
OK	test_verify_recognises_an_untouched_file_and_a_tampered_one		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.50 s
OK	test_version_range_must_be_ordered_and_known		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.76 s

tests/integration/test_releases.py — 6 tests, 6 passed

FR-040-FR-048, BR-006-BR-008.

Result	Test	Purpose	Requirements	Time
OK	test_cannot_change_items_after_publish		BR-006, BR-008, FR-040, FR-048	1.59 s
OK	test_csv_export_of_a_release		BR-006, BR-008, FR-040, FR-048	1.21 s
OK	test_full_database_export_requires_administrator		BR-006, BR-008, FR-040, FR-048	0.63 s
OK	test_full_release_lifecycle		BR-006, BR-008, FR-040, FR-048	1.18 s
OK	test_non_admin_cannot_create_release		BR-006, BR-008, FR-040, FR-048	0.60 s
OK	test_sandbox_revision_rejected_from_release_items		BR-006, BR-008, FR-040, FR-048	0.43 s

tests/integration/test_review_aids.py — 17 tests, 17 passed

I-23 field comments and review checklists, I-13 evidence attachments, I-11 change sets,

Result	Test	Purpose	Requirements	Time
OK	test_attachment_permissions_and_removal_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.82 s
OK	test_attachment_type_size_and_content_checks		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	0.92 s
OK	test_author_attaches_evidence_and_reviewers_download_it		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.18 s
OK	test_auto_assignment_on_submit_is_a_setting		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.12 s
OK	test_bulk_decision_needs_the_password_once_and_a_wrong_one_counts_once		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.34 s
OK	test_bulk_decision_skips_what_the_actor_may_not_decide		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.46 s
OK	test_change_set_groups_requests_and_bulk_decisions_stay_individual		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.50 s
OK	test_change_set_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.81 s
OK	test_checklist_is_administrator_managed_and_inactive_items_are_not_required		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	0.85 s
OK	test_checklist_must_be_confirmed_to_approve_and_is_recorded		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.19 s
OK	test_field_comment_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.24 s
OK	test_least_loaded_reviewer_is_suggested_and_assigned		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.32 s
OK	test_only_the_commenter_author_or_admin_can_resolve		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.63 s
OK	test_out_of_office_reviewers_are_not_auto_assigned		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.18 s
OK	test_rejection_does_not_need_the_checklist		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.11 s
OK	test_removing_a_draft_attachment_deletes_the_file_and_withdrawal_purges_all		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	0.93 s
OK	test_reviewer_comments_on_a_field_and_the_author_sees_and_resolves_it		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.33 s

tests/integration/test_roles_and_languages.py — 40 tests, 40 passed

The StCd Owner role, the role colours, who may create users and hand out roles, and the three

Result	Test	Purpose	Requirements	Time
OK	test_a_string_missing_from_a_catalogue_shows_in_english	Every language is complete (FR-167), but a string that is ever missing must still show.	FR-167	1.29 s
OK	test_arabic_is_right_to_left_but_identifiers_codes_and_numbers_stay_as_they_are		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.97 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ar-\u062a\u0633\u062c\u064a\u0644 \u0627\u0644\u062f\u062e\u0648\u0644]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.73 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[bn-\u09b2\u0997 \u0987\u09a8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[da-Log ind]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[es-Iniciar sesi\xf3n]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.35 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[fr-Se connecter]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.37 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[gu-\u0ab2\u0ac9\u0a97 \u0a87\u0aa8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[hi-\u0932\u0949\u0917 \u0907\u0928 \u0915\u0930\u0947\u0902]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[it-Accedi]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ka-\u10e8\u10d4\u10e1\u10d5\u10da \u10d0]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[kn-\u0cb2\u0cbe\u0c97\u0ccd \u0c87\u0ca8\u0ccd]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ml-\u0d32\u0d4b\u0d17\u0d3f\u0d7b]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.73 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[mr-\u0932\u0949\u0917 \u0907\u0928]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[or-\u0b32\u0b17\u0b4d \u0b07\u0b28\u0b4d]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.35 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[pa-\u0a32\u0a3e\u0a17 \u0a07\u0a28]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ta-\u0b89\u0bb3\u0bcd\u0ba8\u0bc1\ u0bb4\u0bc8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.36 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[te-\u0c32\u0c3e\u0c17\u0c3f\u0c28\u 0c4d]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.35 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ur-\u0644\u0627\u06af \u0627\u0646]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.35 s
OK	test_only_an_administrator_creates_users		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.12 s
OK	test_other_roles_cannot_open_the_user_list		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.89 s

Result	Test	Purpose	Requirements	Time
OK	test_the_administrator_still_gives_every_role		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.76 s
OK	test_the_catalogues_agree_and_only_translate_real_strings		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.18 s
OK	test_the_choice_of_an_added_language_is_remembered_for_the_user		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.90 s
OK	test_the_language_names_have_the_country_letters_left_and_the_flag_right		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.31 s
OK	test_the_last_administrator_is_still_protected		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.16 s
OK	test_the_levels_order_the_roles_and_red_is_the_highest		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.31 s
OK	test_the_menu_shows_letters_left_and_a_drawn_flag_right_for_every_language		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.76 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[admin-VIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.71 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[chief_engineer-REVIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.69 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[engineer-ADMINISTRATOR]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.71 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[reviewer-STCD_OWNER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.72 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[stcd_owner-VIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.70 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[viewer-CHIEF_ENGINEER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.14 s
OK	test_the_stcd_owner_changes_nothing_but_the_role		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.73 s
OK	test_the_stcd_owner_moves_users_between_viewer_engineer_and_reviewer		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.80 s
OK	test_the_stcd_owner_reads_the_workflow_but_changes_nothing_in_it		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.85 s
OK	test_the_stcd_owner_sees_the_list_without_the_administrator_forms		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.90 s
OK	test_the_user_page_colours_every_role_and_red_is_the_administrator		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.79 s
OK	test_who_may_change_which_role_to_which		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.30 s

tests/integration/test_schema_migration.py — 14 tests, 14 passed

US-003/US-004 acceptance criteria: fresh DB matches docs/artifacts/schema.sql

Result	Test	Purpose	Requirements	Time
OK	test_all_documented_tables_exist		US-003, US-004	0.30 s
OK	test_first_subgroup_of_every_main_group_covers_the_first_200_numbers	Rule of 2026-10-22: e.g. Tower 5000-5199, Generator 2000-2199; the next	US-003, US-004	0.31 s
OK	test_foreign_keys_enforced	SEC/BR data-integrity depends on FK enforcement being on for every	US-003, US-004	0.31 s
OK	test_functional_system_group_code_is_reused_for_hub_controller	WTUR and WROT are each intentionally reused for a second,	US-003, US-004	0.31 s
OK	test_hub_sub_bands_are_aligned_to_the_hundreds_of_their_prefix	docs/05-governance-handbook.md section 2: 123xx owns 12300-12399; a band shorter than a	US-003, US-004	0.30 s
OK	test_nineteen_languages_seeded_with_country_letters_left_and_the_flag_right		US-003, US-004	0.31 s
OK	test_no_placeholder_group_is_seeded	Only the 12 real groups exist out of the box; free ranges stay free.	US-003, US-004	0.30 s
OK	test_only_sandbox_status_codes_can_be_deleted	Exercises trg_prevent_nonsandbox_delete from docs/artifacts/schema.sql.	US-003, US-004	0.32 s
OK	test_second_migration_is_reversible_and_idempotent	A database already at the earlier baseline is brought up to date by the	US-003, US-004	0.93 s

Result	Test	Purpose	Requirements	Time
OK	test_six_roles_seeded		US-003, US-004	0.30 s
OK	test_subgroup_codes_have_no_dash_and_no_letter_after_the_x		US-003, US-004	0.30 s
OK	test_thirty_eight_functional_subgroups_seeded		US-003, US-004	0.31 s
OK	test_three_turbine_platforms_seeded		US-003, US-004	0.71 s
OK	test_twelve_functional_system_groups_seeded	9 Main Controller groups + 3 Hub Controller groups (WTUR/WROT/WPPD	US-003, US-004	0.31 s

tests/integration/test_schema_upgrade.py — 16 tests, 16 passed

Upgrading a database that was created by the previous release (Alembic d6f8b0c2e4a3, schema v3)

Result	Test	Purpose	Requirements	Time
OK	test_a_second_upgrade_is_a_no_op_and_the_downgrade_returns_to_v3_with_the_data		AUD-074, BR-016, FR-147, FR-148, TC-045	1.39 s
OK	test_an_upgraded_database_has_the_same_shape_as_a_fresh_one		AUD-074, BR-016, FR-147, FR-148, TC-045	1.04 s
OK	test_covert_is_not_visible_and_the_other_audiences_still_are		AUD-074, BR-016, FR-147, FR-148, TC-045	0.70 s
OK	test_every_trigger_of_a_fresh_database_exists_after_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	1.02 s
OK	test_existing_data_survives_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	0.72 s
OK	test_hub_groups_become_hub_controller_groups		AUD-074, BR-016, FR-147, FR-148, TC-045	0.73 s
OK	test_hub_sub_bands_are_shifted_to_the_hundreds_by_the_upgrade	The v3 database still has the legacy x001 bands; the upgrade aligns them.	AUD-074, BR-016, FR-147, FR-148, TC-045	0.73 s
OK	test_import_job_is_rebuilt_for_programs_without_losing_rows_or_references	v4.2: the CHECK of import_job.source_format has to widen to 'ST', which SQLite can only do by	AUD-074, BR-016, FR-147, FR-148, TC-045	17.67 s
OK	test_new_columns_default_sensibly_on_old_rows		AUD-074, BR-016, FR-147, FR-148, TC-045	1.11 s
OK	test_old_audit_rows_are_chained_and_the_chain_verifies		AUD-074, BR-016, FR-147, FR-148, TC-045	17.13 s
OK	test_the_languages_and_export_profiles_are_seeded		AUD-074, BR-016, FR-147, FR-148, TC-045	0.73 s
OK	test_the_migrated_hash_equals_what_the_application_would_have_written	Migration and application share one canonical form (app/audit_hash.py).	AUD-074, BR-016, FR-147, FR-148, TC-045	1.12 s
OK	test_the_search_index_is_filled_with_the_old_codes		AUD-074, BR-016, FR-147, FR-148, TC-045	0.73 s
OK	test_the_subgroup_codes_lose_their_letters_and_dashes_in_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	0.71 s
OK	test_the_table_rebuilds_leave_every_foreign_key_intact	import_job (revision b3f5a7c9d124) and user (d4a6c8e0f236) are rebuilt to widen a CHECK; the	AUD-074, BR-016, FR-147, FR-148, TC-045	17.09 s
OK	test_the_upgraded_database_works_for_a_whole_workflow		AUD-074, BR-016, FR-147, FR-148, TC-045	0.91 s

tests/integration/test_search_and_help.py — 16 tests, 16 passed

I-31 ranked, typo-tolerant search (with a substring fallback), I-35 in-app help content.

Result	Test	Purpose	Requirements	Time
OK	test_a_hit_in_the_title_outranks_a_hit_in_the_description		FR-148, FR-154, TC-047	0.82 s
OK	test_a_new_code_and_an_edit_are_reflected_without_a_rebuild		FR-148, FR-154, TC-047	0.88 s
OK	test_every_lifecycle_state_role_and_cr_state_has_help		FR-148, FR-154, TC-047	0.00 s
OK	test_help_texts_are_plain_sentences		FR-148, FR-154, TC-047	0.00 s
OK	test_new_terms_are_in_the_project_glossary		FR-148, FR-154, TC-047	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_prefix_and_multi_word_search		FR-148, FR-154, TC-047	0.83 s
OK	test_search_ignores_fts_operators_and_punctuation		FR-148, FR-154, TC-047	1.44 s
OK	test_several_misspelt_words_are_all_tolerated_together	Each word may expand to a group of close spellings; the groups must be joined with AND	FR-148, FR-154, TC-047	0.84 s
OK	test_state_badges_carry_their_glossary_text_as_a_tooltip		FR-148, FR-154, TC-047	0.90 s
OK	test_the_change_request_screen_says_what_happens_next		FR-148, FR-154, TC-047	1.51 s
OK	test_the_help_page_shows_the_quick_start_of_the_signed_in_role		FR-148, FR-154, TC-047	1.04 s
OK	test_the_roles_and_states_are_named_in_the_project_glossary		FR-148, FR-154, TC-047	0.00 s
OK	test_the_search_box_works_on_the_ui_list		FR-148, FR-154, TC-047	0.92 s
OK	test_the_search_index_exists_on_a_fresh_database		FR-148, FR-154, TC-047	16.75 s
OK	test_typos_still_find_the_code		FR-148, FR-154, TC-047	0.81 s
OK	test_without_the_index_search_falls_back_to_substrings		FR-148, FR-154, TC-047	0.91 s

tests/integration/test_select_all.py — 2 tests, 2 passed

A list with a checkbox per row has a checkbox in its header that ticks them all (the helper is

Result	Test	Purpose	Requirements	Time
OK	test_every_list_with_a_checkbox_per_row_can_be_selected_as_a_whole		FR-159, FR-180, NFR-007, TC-036, TC-070	0.14 s
OK	test_the_header_box_and_the_row_boxes_share_their_class		FR-159, FR-180, NFR-007, TC-036, TC-070	0.01 s

tests/integration/test_sla_and_notifications.py — 16 tests, 16 passed

I-08 SLA ages, stalled requests, in-app notices and reminders (docs/21-improvement-ideas.md),

Result	Test	Purpose	Requirements	Time
OK	test_assigned_reviewer_only_and_pending_approval_goes_to_chief_engineers		FR-120, FR-121, FR-122, FR-123, TC-041	2.03 s
OK	test_business_days_skip_weekends		FR-120, FR-121, FR-122, FR-123, TC-041	0.00 s
OK	test_kpis_are_computed_from_stored_timestamps		FR-120, FR-121, FR-122, FR-123, TC-041	1.91 s
OK	test_kpis_flag_an_incomplete_audit_trail_and_an_ownerless_code		FR-120, FR-121, FR-122, FR-123, TC-041	1.31 s
OK	test_kpis_without_data_do_not_divide_by_zero		FR-120, FR-121, FR-122, FR-123, TC-041	0.32 s
OK	test_levels_follow_the_thresholds		FR-120, FR-121, FR-122, FR-123, TC-041	0.92 s
OK	test_mail_failure_never_breaks_the_reminder_job		FR-120, FR-121, FR-122, FR-123, TC-041	1.01 s
OK	test_mark_all_read		FR-120, FR-121, FR-122, FR-123, TC-041	1.00 s
OK	test_queues_show_the_waiting_time		FR-120, FR-121, FR-122, FR-123, TC-041	0.98 s
OK	test_reminders_also_reach_a_covering_delegate_and_can_be_mailed		FR-120, FR-121, FR-122, FR-123, TC-041	1.15 s
OK	test_reminders_go_to_whoever_holds_the_next_decision_once_a_day		FR-120, FR-121, FR-122, FR-123, TC-041	1.30 s
OK	test_send_reminders_cli		FR-120, FR-121, FR-122, FR-123, TC-041	0.32 s
OK	test_stalled_view_and_reassignment		FR-120, FR-121, FR-122, FR-123, TC-041	1.28 s
OK	test_the_default_thresholds_are_the_documented_ones		FR-120, FR-121, FR-122, FR-123, TC-041	0.32 s
OK	test_thresholds_are_configurable_and_validated		FR-120, FR-121, FR-122, FR-123, TC-041	0.80 s
OK	test_withdrawn_cr_notices_are_removed_with_it		FR-120, FR-121, FR-122, FR-123, TC-041	0.91 s

tests/integration/test_st_category_and_draft_delete.py — 8 tests, 8 passed

The status category per program of a .st import, and the bulk delete of Drafts on the Status

Result	Test	Purpose	Requirements	Time
OK	test_a_category_chosen_for_one_program_wins		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	0.89 s
OK	test_a_type_without_a_choice_gets_the_default_category		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	0.79 s
OK	test_all_own_drafts_of_a_filter_are_deleted_on_every_page		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	1.03 s
OK	test_an_unknown_category_is_refused		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	0.79 s
OK	test_somebody_elses_drafts_cannot_be_ticked_or_deleted		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	1.11 s
OK	test_the_alarm_type_of_a_program_sets_its_category		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	0.89 s
OK	test_the_list_offers_tick_boxes_for_the_own_drafts_and_deletes_the_ticked		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	1.02 s
OK	test_the_preview_page_lets_the_user_set_the_category_per_program		FR-026, FR-159, FR-181, FR-182, TC-053, TC-070	1.46 s

tests/integration/test_st_import.py — 29 tests, 29 passed

Status code import from the generated StCd.st programs and the legacy -> new mapping_

Result	Test	Purpose	Requirements	Time
OK	test_a_file_without_programs_or_too_large_is_refused_cleanly		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.00 s
OK	test_a_hub_program_keeps_its_legacy_number_and_gets_the_new_one_from_the_table		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.01 s
OK	test_a_legacy_hub_number_missing_from_the_table_leaves_no_identifier		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.00 s
OK	test_a_legacy_number_that_belongs_to_another_code_blocks_the_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.59 s
OK	test_a_number_defined_twice_in_one_upload_blocks_the_second		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.45 s
OK	test_a_preview_belongs_to_its_uploader_and_expires		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.66 s
OK	test_a_second_preview_shows_what_is_already_there_and_confirm_refuses_it		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.19 s
OK	test_a_used_prg_idx_is_a_warning_and_is_not_stored_twice		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.56 s
OK	test_a_zip_and_loose_files_are_read_alike_and_other_files_are_noted		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.48 s
OK	test_a_zip_that_unpacks_too_large_is_refused		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.46 s
OK	test_allocation_skips_a_reserved_number_and_a_manual_override_cannot_take_it		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.50 s
OK	test_an_administrator_cannot_override_a_reserved_identifier		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.98 s
OK	test_an_expired_preview_is_refused_and_removed		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.47 s
OK	test_confirm_creates_drafts_that_reserve_their_numbers_in_one_change_set		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.57 s
OK	test_confirming_without_a_selection_takes_every_ready_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.55 s
OK	test_nothing_usable_is_a_clear_error		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.45 s
OK	test_only_engineers_and_administrators_import		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.44 s
OK	test_proposal_two_places_the_hub_programs_inside_the_main_scheme		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.46 s
OK	test_submitting_issues_exactly_the_reserved_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.00 s
OK	test_the_bundled_hub_table_equals_the_workbook		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.04 s

Result	Test	Purpose	Requirements	Time
OK	test_the_controller_can_be_forced_for_a_file		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.88 s
OK	test_the_database_refuses_a_second_reservation_of_a_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.52 s
OK	test_the_draft_carries_the_attributes_of_the_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.50 s
OK	test_the_mapping_exports_as_csv_and_xlsx		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.57 s
OK	test_the_mapping_is_not_for_viewers		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.44 s
OK	test_the_mapping_lists_legacy_to_new_and_counts_codes_without_a_legacy_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.13 s
OK	test_the_preview_lists_every_program_and_creates_nothing		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.50 s
OK	test_the_reader_takes_the_attributes_of_a_main_controller_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.01 s
OK	test_the_source_of_an_imported_code_can_be_read		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.51 s

tests/integration/test_status_codes.py — 9 tests, 9 passed

FR-020-FR-029, FR-053-FR-056, FR-060-FR-064.

Result	Test	Purpose	Requirements	Time
OK	test_admin_sandbox_gets_identifier_immediately		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.45 s
OK	test_create_rejects_invalid_fields		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.95 s
OK	test_create_rejects_missing_functional_subgroup	BR-013: mandatory, never optional (corrected 2026-09-17).	BR-013	0.45 s
OK	test_engineer_can_create_draft_with_no_identifier		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.46 s
OK	test_non_admin_cannot_create_sandbox		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.45 s
OK	test_sandbox_delete_requires_admin_and_sandbox_flag		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.67 s
OK	test_search_defaults_exclude_sandbox		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.48 s
OK	test_viewer_cannot_create_status_code		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.43 s
OK	test_viewer_cannot_see_draft_status_codes		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.59 s

tests/integration/test_step_up_and_delegation.py — 16 tests, 16 passed

I-17 step-up re-authentication (SEC-026) and I-16 delegation / out-of-office cover.

Result	Test	Purpose	Requirements	Time
OK	test_admin_signoff_chief_decision_publish_and_full_export_all_need_the_password		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.57 s
OK	test_correct_password_lets_the_signoff_through_and_is_audited		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.08 s
OK	test_deactivated_delegator_ends_the_delegation		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.08 s
OK	test_delegate_can_sign_off_as_the_absent_reviewer_and_audit_shows_both		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.12 s
OK	test_delegate_sees_the_review_queue_and_a_delegated_chief_engineer_can_approve		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	2.27 s
OK	test_delegating_does_not_grant_other_privileges		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	0.57 s

Result	Test	Purpose	Requirements	Time
OK	test_delegation_does_not_lift_segregation_of_duties	The delegate acts as themselves: the author cannot sign their own CR through a delegation.	BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.04 s
OK	test_delegation_is_limited_to_its_period_and_can_be_revoked		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.16 s
OK	test_delegation_validation_and_who_may_create_it		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	0.59 s
OK	test_one_person_cannot_give_both_review_signoffs_through_a_delegation		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.20 s
OK	test_repeated_wrong_passwords_lock_the_account	The prompt must not be usable to guess a password (same lockout as login).	BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.46 s
OK	test_reviewer_signoff_without_password_is_refused		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.45 s
OK	test_step_up_can_be_switched_off_by_configuration		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.04 s
OK	test_ui_decision_forms_carry_a_password_field		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.71 s
OK	test_ui_signoff_without_password_shows_an_error		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.25 s
OK	test_wrong_password_is_refused_and_audited		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.07 s

tests/integration/test_terminology.py — 13 tests, 13 passed

The technical nomenclature: a vocabulary of its own, apart from the interface translations

Result	Test	Purpose	Requirements	Time
OK	test_a_formula_in_a_cell_is_neutralised_in_the_csv		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.93 s
OK	test_a_proposal_is_a_draft_until_another_person_approves_it		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.38 s
OK	test_a_shipped_draft_can_be_approved_by_anybody_who_may_approve_and_all_at_once		AUD-077, BR-018, FR-166, SEC-010, TC-061	8.24 s
OK	test_an_interface_reviewer_has_no_say_in_the_terminology		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.81 s
OK	test_everybody_reads_but_only_the_technical_roles_propose		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.73 s
OK	test_only_the_managers_add_and_change_terms_and_a_new_english_reopens_approvals		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.93 s
OK	test_send_back_needs_a_note_and_reopens_an_approved_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	7.51 s
OK	test_terms_that_are_never_translated_have_no_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.32 s
OK	test_the_check_finds_interface_strings_that_do_not_use_the_term		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.40 s
OK	test_the_shipped_terms_agree_with_the_interface_catalogue_where_both_have_the_word	Role names are interface strings and terms: they must be the same word.	AUD-077, BR-018, FR-166, SEC-010, TC-061	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_the_vocabulary_is_exported_per_language_and_for_all		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.23 s
OK	test_the_vocabulary_ships_with_every_term_in_every_language_as_a_draft		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.35 s
OK	test_validation_of_a_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.86 s

tests/integration/test_translation_approval.py — 12 tests, 12 passed

New and changed translations are approved separately, and every change of the text or of the

Result	Test	Purpose	Requirements	Time
OK	test_a_change_of_the_nomenclature_needs_the_flag_as_well		FR-164, FR-170, FR-172, FR-176, TC-067	2.59 s
OK	test_a_changed_string_must_get_a_verdict_before_the_record_is_current		FR-164, FR-170, FR-172, FR-176, TC-067	2.64 s
OK	test_a_delta_read_one_by_one_can_lower_or_keep_the_confidence		FR-164, FR-170, FR-172, FR-176, TC-067	1.46 s
OK	test_a_follow_up_review_covers_only_what_is_new_or_changed		FR-164, FR-170, FR-172, FR-176, TC-067	1.58 s
OK	test_a_language_never_reviewed_stays_hidden_from_a_viewer		FR-164, FR-170, FR-172, FR-176, TC-067	0.84 s
OK	test_a_reviewed_language_has_nothing_pending_until_a_string_changes		FR-164, FR-170, FR-172, FR-176, TC-067	0.89 s
OK	test_a_viewer_reads_a_changed_string_in_english_until_it_is_approved		FR-164, FR-170, FR-172, FR-176, TC-067	1.18 s
OK	test_an_accepted_proposal_is_approved_and_an_undecided_one_is_not		FR-164, FR-170, FR-172, FR-176, TC-067	0.87 s
OK	test_every_change_of_the_text_or_the_nomenclature_has_been_through_the_check	The gate: this fails while a string or a term changed and the check was not done for it.	FR-164, FR-170, FR-172, FR-176, TC-067	0.19 s
OK	test_the_colour_intervals_run_from_red_to_green		FR-164, FR-170, FR-172, FR-176, TC-067	0.00 s
OK	test_the_overview_shows_the_confidence_of_every_language_in_colour		FR-164, FR-170, FR-172, FR-176, TC-067	4.17 s
OK	test_the_record_holds_a_confidence_for_every_language_with_its_findings		FR-164, FR-170, FR-172, FR-176, TC-067	0.00 s

tests/integration/test_translation_impact.py — 11 tests, 11 passed

A change of the vocabulary is tested automatically and its language impact reaches the StCd Owner

Result	Test	Purpose	Requirements	Time
OK	test_a_changed_nomenclature_is_named_in_the_notice		FR-166, FR-176, FR-177, FR-178, TC-068	0.72 s
OK	test_a_changed_term_and_an_approved_translation_are_reported_again		FR-166, FR-176, FR-177, FR-178, TC-068	0.92 s
OK	test_a_failing_notice_never_undoes_the_change		FR-166, FR-176, FR-177, FR-178, TC-068	0.69 s
OK	test_a_new_term_tells_the_stcd_owner_which_languages_need_attention		FR-166, FR-176, FR-177, FR-178, TC-068	0.82 s
OK	test_a_term_that_stays_in_english_touches_no_language		FR-166, FR-176, FR-177, FR-178, TC-068	0.80 s
OK	test_changed_shipped_text_is_reported_once_per_change		FR-166, FR-176, FR-177, FR-178, TC-068	1.08 s
OK	test_nothing_is_reported_while_the_confidence_check_is_current		FR-166, FR-176, FR-177, FR-178, TC-068	1.05 s
OK	test_the_command_for_the_deployment_reports_and_notifies		FR-166, FR-176, FR-177, FR-178, TC-068	0.70 s
OK	test_the_notice_appears_on_the_notices_page_of_the_stcd_owner		FR-166, FR-176, FR-177, FR-178, TC-068	0.89 s
OK	test_the_reviewer_of_an_affected_language_is_told_too		FR-166, FR-176, FR-177, FR-178, TC-068	0.81 s
OK	test_the_start_of_the_server_and_the_installers_run_the_check		FR-166, FR-176, FR-177, FR-178, TC-068	0.02 s

tests/integration/test_translation_review.py — 14 tests, 14 passed

Translation review in the application (docs/26-translations.md, docs/03-srs.md FR-164): native

Result	Test	Purpose	Requirements	Time
OK	test_a_missing_translation_can_be_proposed_and_accepted		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	7.76 s
OK	test_a_proposal_awaits_approval_until_it_is_decided		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	12.01 s
OK	test_a_proposal_must_keep_the_placeholders_and_an_unclear_string_is_only_noted		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	28.69 s
OK	test_a_review_is_completed_only_when_every_string_has_a_verdict_and_is_recorded		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	9.14 s
OK	test_a_reviewer_answers_strings_and_is_told_what_is_wrong		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	8.71 s
OK	test_a_sheet_that_is_not_one_is_refused_clearly		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	8.17 s
OK	test_an_accepted_proposal_is_shown_at_once_and_can_be_taken_back		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	23.05 s
OK	test_an_account_that_reviews_has_history_in_the_usage_report		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	8.73 s
OK	test_an_unassigned_user_cannot_open_a_review_and_an_unknown_language_is_404		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	0.92 s
OK	test_nobody_decides_on_their_own_proposal_and_a_decision_is_final		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	32.69 s
OK	test_only_an_administrator_or_an_stcd_owner_assigns_reviewers_and_it_is_audited		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	23.79 s
OK	test_right_to_left_languages_are_shown_right_to_left_in_the_review		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	17.23 s
OK	test_the_overview_shows_a_reviewer_only_their_languages_and_a_manager_all		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	18.62 s
OK	test_the_sheet_can_be_downloaded_filled_in_offline_and_uploaded		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	11.58 s

tests/integration/test_ui_screens.py — 18 tests, 18 passed

S1–S19 server-rendered UI: every screen renders for the roles that can

Result	Test	Purpose	Requirements	Time
OK	test_admin_signoff_form_offers_access_override_and_submit_override		NFR-007a, TC-036, UAT-13	1.32 s
OK	test_comments_api_validates_and_is_role_restricted		NFR-007a, TC-036, UAT-13	1.08 s
OK	test_cr_detail_shows_role_specific_controls		NFR-007a, TC-036, UAT-13	1.12 s
OK	test_default_role_for_new_user_is_viewer	SEC-012.	SEC-012	0.47 s
OK	test_draft_cr_page_says_it_must_still_be_submitted		NFR-007a, TC-036, UAT-13	1.48 s
OK	test_every_screen_renders_for_an_administrator		NFR-007a, TC-036, UAT-13	1.97 s
OK	test_failed_form_submission_keeps_every_entered_value	NFR-007a / TC-036 / UAT-13.	NFR-007a, TC-036, UAT-13	0.94 s
OK	test_lookup_description_is_editable_and_flows_to_form_and_api		NFR-007a, TC-036, UAT-13	1.03 s
OK	test_lookup_selects_carry_inline_help_hooks	NFR-007b: each lookup-backed select has a help target and per-option text.	NFR-007b	1.41 s
OK	test_people_are_shown_by_name_and_lists_show_the_title	Authors, signers and commenters appear as "Full Name (Role)", never "User #n";	NFR-007a, TC-036, UAT-13	1.99 s
OK	test_role_specific_screens_render[chief_engineer-/change-requests/approval-queue]		NFR-007a, TC-036, UAT-13	0.83 s
OK	test_role_specific_screens_render[engineer-/change-requests/mine]		NFR-007a, TC-036, UAT-13	0.80 s
OK	test_role_specific_screens_render[engineer-/exports]		NFR-007a, TC-036, UAT-13	0.77 s

Result	Test	Purpose	Requirements	Time
OK	test_role_specific_screens_render[reviewer-/change-requests/review-queue]		NFR-007a, TC-036, UAT-13	0.83 s
OK	test_role_specific_screens_render[viewer-/exports]		NFR-007a, TC-036, UAT-13	0.84 s
OK	test_role_specific_screens_render[viewer-/status-codes]		NFR-007a, TC-036, UAT-13	1.38 s
OK	test_sign_off_request_listed_on_my_change_requests		NFR-007a, TC-036, UAT-13	1.04 s
OK	test_ui_comment_reviewer_assignment_and_override_flow		NFR-007a, TC-036, UAT-13	1.56 s

tests/integration/test_user_administration.py — 10 tests, 10 passed

Viewing, deactivating and deleting users (docs/03-srs.md FR-158, BR-025, SEC-032). An account that

Result	Test	Purpose	Requirements	Time
OK	test_a_deactivated_account_can_be_activated_again		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.76 s
OK	test_an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.93 s
OK	test_an_account_with_history_is_erased_and_the_audit_log_is_untouched		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.58 s
OK	test_deleting_needs_the_administrators_password_and_an_administrator		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.32 s
OK	test_nobody_deletes_their_own_account		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.75 s
OK	test_the_buttons_deactivate_activate_and_delete_through_the_screens		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.10 s
OK	test_the_detail_page_shows_history_and_the_buttons_follow_it		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.07 s
OK	test_the_stcd_owner_sees_view_and_set_role_but_no_delete_or_deactivate		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.88 s
OK	test_the_user_list_offers_view_edit_deactivate_and_delete_to_an_administrator		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.83 s
OK	test_usage_is_open_to_the_user_list_roles_only		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	0.90 s

tests/integration/test_users.py — 7 tests, 7 passed

FR-010-FR-014, SEC-003 §3 permission matrix, US-009.

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_force_reset_another_users_password		FR-010, FR-014, SEC-003, US-009	0.60 s
OK	test_admin_can_list_and_create_users		FR-010, FR-014, SEC-003, US-009	0.46 s
OK	test_can_deactivate_administrator_when_another_remains_active		FR-010, FR-014, SEC-003, US-009	0.46 s
OK	test_cannot_deactivate_last_active_administrator		FR-010, FR-014, SEC-003, US-009	0.40 s
OK	test_create_user_rejects_weak_password		FR-010, FR-014, SEC-003, US-009	0.41 s
OK	test_non_admin_cannot_list_users		FR-010, FR-014, SEC-003, US-009	0.97 s
OK	test_self_password_change		FR-010, FR-014, SEC-003, US-009	0.66 s

tests/integration/test_withdraw_and_reimport.py — 11 tests, 11 passed

Withdrawing change requests (several at once) and what a withdrawal does to a program that was

Result	Test	Purpose	Requirements	Time
OK	test_a_change_request_of_somebody_else_is_refused_and_the_others_go_on		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	0.93 s
OK	test_a_changed_program_opens_a_new_revision_of_an_approved_code		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.49 s
OK	test_a_changed_program_updates_the_open_draft_when_it_is_ticked		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.50 s

Result	Test	Purpose	Requirements	Time
OK	test_a_restored_code_is_no_shell_and_the_database_refuses_a_delete_around_the_service		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.16 s
OK	test_a_withdrawn_code_is_listed_and_an_administrator_deletes_it_and_imports_again		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.33 s
OK	test_a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.15 s
OK	test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	0.94 s
OK	test_bulk_withdraw_needs_a_list_and_a_role_that_can_withdraw		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.40 s
OK	test_several_change_requests_are_withdrawn_in_one_go_and_each_is_reported		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	0.76 s
OK	test_the_list_has_tick_boxes_and_a_withdraw_selected_button		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.03 s
OK	test_the_preview_screen_lists_the_choices_and_leaves_changed_programs_unticked		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.18 s

tests/integration/test_workflow_gaps.py — 18 tests, 18 passed

Closes the gaps found in the spec-vs-code review (docs/20-traceability-matrix.md):

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_override_identifier_with_reason_and_it_is_audited		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.88 s
OK	test_admin_signoff_can_override_proposed_audience_access		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.21 s
OK	test_assigned_reviewer_only_may_sign_off		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.15 s
OK	test_cr_page_explains_when_no_engineer_holds_a_pending_domain		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.45 s
OK	test_domain_request_is_refused_when_nobody_holds_the_domain	A request nobody could ever act on must not be silently accepted.	BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.60 s
OK	test_domain_signoff_request_surfaces_for_target_and_clears_once_signed		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.12 s
OK	test_domain_signoff_request_validation		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.94 s
OK	test_identifier_override_forbidden_for_engineer		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.84 s
OK	test_identifier_override_rejects_out_of_range_malformed_and_duplicate		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.76 s
OK	test_identifier_override_requires_reason		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.86 s
OK	test_no_reassignment_after_reviewer_signed_off		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.05 s
OK	test_only_administrator_endpoint_accepts_overrides		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.91 s
OK	test_rejection_ignores_access_overrides		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.08 s
OK	test_reviewer_assignment_validation_and_authorization		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.47 s
OK	test_reviewer_can_be_reassigned_and_unassigned		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.22 s
OK	test_unknown_access_override_field_is_rejected		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.96 s
OK	test_untargeted_domain_request_is_visible_to_every_domain_holder		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.94 s

Result	Test	Purpose	Requirements	Time
OK	test_withdrawn_cr_removes_its_domain_requests		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.96 s

tests/system/test_accessibility.py — 9 tests, 9 passed

I-22 accessibility (WCAG 2.1 AA, docs/10-ui-ux-specification.md §12): colour contrast of the

Result	Test	Purpose	Requirements	Time
OK	test_every_page_is_structurally_accessible[de]		FR-155, NFR-010, TC-049	2.17 s
OK	test_every_page_is_structurally_accessible[en]		FR-155, NFR-010, TC-049	2.35 s
OK	test_every_state_badge_meets_aa		FR-155, NFR-010, TC-049	0.00 s
OK	test_status_is_never_conveyed_by_colour_alone	Every state badge carries the state as text, not only a colour.	FR-155, NFR-010, TC-049	0.87 s
OK	test_text_tokens_meet_aa_on_their_backgrounds		FR-155, NFR-010, TC-049	0.00 s
OK	test_the_chief_engineers_queue_is_accessible		FR-155, NFR-010, TC-049	0.77 s
OK	test_the_contrast_helper_matches_the_wcag_reference_values		FR-155, NFR-010, TC-049	0.01 s
OK	test_the_login_page_is_accessible		FR-155, NFR-010, TC-049	0.36 s
OK	test_the_stylesheet_honours_user_preferences		FR-155, NFR-010, TC-049	0.00 s

tests/system/test_cli_create_admin.py — 2 tests, 2 passed

Result	Test	Purpose	Requirements	Time
OK	test_create_admin_cli_command			0.92 s
OK	test_create_admin_cli_rejects_weak_password			0.31 s

tests/system/test_dark_mode.py — 8 tests, 8 passed

Dark mode (docs/10-ui-ux-specification.md section 13, FR-183): the dark palette meets the same

Result	Test	Purpose	Requirements	Time
OK	test_colours_are_only_written_out_where_the_dark_theme_cannot_matter	Outside the token blocks a hex colour is allowed only for fills that carry their own text	FR-155, FR-183, NFR-010, SEC-025, TC-049	0.01 s
OK	test_every_badge_and_sla_pill_meets_aa_in_the_dark_theme		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_palette_meets_aa_on_its_backgrounds		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_theme_gives_every_light_tint_a_dark_value	A token added to the light palette without a dark twin would stay light on a dark page.	FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_theme_is_for_the_screen_only		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_login_page_and_every_signed_in_page_offer_the_theme_choice		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.94 s
OK	test_the_theme_choice_is_translated		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.80 s
OK	test_theme_js_is_a_plain_file_that_follows_the_system_and_remembers_the_choice		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s

tests/system/test_deploy_artifacts.py — 22 tests, 22 passed

I-33 / I-34: the packaging, installer, scheduler units and CI definitions are consistent with

Result	Test	Purpose	Requirements	Time
OK	test_backup_runs_every_night_and_reminders_on_weekdays		NFR-013, NFR-014, TC-051	0.00 s
OK	test_ci_files_parse_and_run_the_ci_script		NFR-013, NFR-014, TC-051	0.01 s
OK	test_create_admin_can_take_its_answers_from_the_environment		NFR-013, NFR-014, TC-051	0.37 s
OK	test_each_timer_has_a_matching_oneshot_service		NFR-013, NFR-014, TC-051	0.01 s
OK	test_every_expected_artifact_is_shipped		NFR-013, NFR-014, TC-051	0.01 s
OK	test_every_pattern_in_the_scan_is_classified		NFR-013, NFR-014, TC-051	0.00 s
OK	test_every_setting_an_installer_writes_is_read_by_the_application		NFR-013, NFR-014, TC-051	0.01 s
OK	test_every_setting_in_the_example_file_is_read_by_the_application		NFR-013, NFR-014, TC-051	0.01 s
OK	test_linux_installer_parses_and_its_dry_run_changes_nothing		NFR-013, NFR-014, TC-051	1.38 s
OK	test_non_sqlite_connections_are_left_alone	The engine listeners must not touch another driver's connection (docs/07 section 12).	NFR-013, NFR-014, TC-051	0.31 s
OK	test_production_refuses_to_start_with_a_weak_key		NFR-013, NFR-014, TC-051	4.31 s
OK	test_serve_options_follow_the_environment		NFR-013, NFR-014, TC-051	0.02 s
OK	test_services_call_only_commands_that_exist		NFR-013, NFR-014, TC-051	0.32 s
OK	test_the_ci_script_documents_itself		NFR-013, NFR-014, TC-051	1.13 s
OK	test_the_postgres_ratchet_holds		NFR-013, NFR-014, TC-051	5.53 s
OK	test_the_production_entry_point_serves_and_reports_its_state		NFR-013, NFR-014, TC-051	10.12 s
OK	test_the_ratchet_notices_new_sqlite_specific_code		NFR-013, NFR-014, TC-051	5.82 s
OK	test_units_parse_and_have_the_required_sections		NFR-013, NFR-014, TC-051	0.02 s
OK	test_web_service_runs_the_production_entry_point_as_an_unprivileged_user		NFR-013, NFR-014, TC-051	0.00 s
OK	test_windows_installer_dry_run_lists_every_step_and_creates_nothing		NFR-013, NFR-014, TC-051	2.26 s
OK	test_windows_scripts_parse		NFR-013, NFR-014, TC-051	2.41 s
OK	test_write_paths_of_the_units_cover_what_the_jobs_write		NFR-013, NFR-014, TC-051	0.00 s

tests/system/test_docs_consistency.py — 17 tests, 17 passed

TC-052. The documents agree with the code and with each other: the generated endpoint table, the

Result	Test	Purpose	Requirements	Time
OK	test_every_requirement_id_cited_in_a_document_is_defined		TC-052	0.43 s
OK	test_every_requirement_is_accounted_for_and_the_matrix_evidence_resolves		TC-052	1.12 s
OK	test_every_screen_path_in_the_ui_inventory_is_a_real_route	docs/10 section 1 names the path of each v4 screen; each must exist as a UI route.	TC-052	0.30 s
OK	test_every_test_case_id_cited_in_the_documents_exists		TC-052	0.08 s
OK	test_links_between_documents_point_at_existing_files_and_headings		TC-052	0.14 s
OK	test_paragraphs_in_documents_are_single_source_lines	docs/15 §6: prose paragraphs are one continuous source line (tables, lists and code may span).	TC-052	0.06 s
OK	test_section_references_between_documents_exist		TC-052	0.18 s
OK	test_the_audit_catalogue_lists_every_code_the_application_can_write		TC-052	0.47 s
OK	test_the_contract_version_and_operation_count_quoted_in_the_documents		TC-052	0.50 s

Result	Test	Purpose	Requirements	Time
OK	test_the_decisions_and_assumptions_are_complete		TC-052	0.03 s
OK	test_the_document_set_version_is_current_and_the_report_is_generated		TC-052	0.01 s
OK	test_the_endpoint_table_of_the_api_document_is_the_generated_one		TC-052	0.62 s
OK	test_the_group_and_subgroup_counts_quoted_in_the_documents		TC-052	0.00 s
OK	test_the_screen_count_matches_the_inventory		TC-052	0.01 s
OK	test_the_table_and_trigger_counts_quoted_in_the_documents		TC-052	0.03 s
OK	test_the_test_suite_table_of_the_strategy_matches_the_test_files	docs/13 section 8a quotes how many test functions each v4 module has.	TC-052	0.08 s
OK	test_the_v4_tables_of_the_data_dictionary_name_real_columns	docs/06 section 13 lists each v4 table with its fields; every named field must exist.	TC-052	0.01 s

tests/system/test_docs_pdf.py — 34 tests, 34 passed

The documents of a release as PDF with the version in the file name (FR-185,

Result	Test	Purpose	Requirements	Time
OK	test_a_document_is_built_with_the_version_in_its_name		FR-184, FR-185, NFR-013, TC-072	0.32 s
OK	test_an_unknown_document_is_refused		FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_every_character_of_the_documents_can_be_drawn[00-INDEX.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[01-executive-summary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_every_character_of_the_documents_can_be_drawn[02-prd.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[03-srs.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.02 s
OK	test_every_character_of_the_documents_can_be_drawn[04-domain-model.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[05-governance-handbook.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[06-data-dictionary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[07-database-design.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[08-system-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[09-api-specification.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[10-ui-ux-specification.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[11-security-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[12-audit-compliance.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[13-test-strategy.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[14-deployment-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[15-development-standards.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_every_character_of_the_documents_can_be_drawn[16-mvp-roadmap.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_every_character_of_the_documents_can_be_drawn[17-implementation-backlog.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s

Result	Test	Purpose	Requirements	Time
OK	test_every_character_of_the_documents_can_be_drawn[18-claude-code-implementation-guide.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[19-glossary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[20-traceability-matrix.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[21-improvement-ideas.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[22-software-description.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[23-user-manual.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[24-test-report.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.03 s
OK	test_every_character_of_the_documents_can_be_drawn[25-hc-integration-concept.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_inline_markup_is_removed_and_kept_as_styles		FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_symbols_a_pdf_font_cannot_draw_become_words		FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_the_four_documents_have_their_file_names		FR-184, FR-185, NFR-013, TC-072	0.07 s
OK	test_the_full_documentation_has_a_contents_list_and_the_artefacts		FR-184, FR-185, NFR-013, TC-072	0.14 s
OK	test_the_reader_finds_every_kind_of_block		FR-184, FR-185, NFR-013, TC-072	0.04 s
OK	test_the_user_manual_is_built_from_the_real_document		FR-184, FR-185, NFR-013, TC-072	3.10 s

tests/system/test_first_start.py — 3 tests, 3 passed

The first start on a new machine is one command that works from any folder

Result	Test	Purpose	Requirements	Time
OK	test_every_step_is_listed_from_any_folder_and_nothing_is_changed		NFR-013, TC-071	0.60 s
OK	test_the_port_and_the_skipped_administrator_are_honoured		NFR-013, TC-071	0.47 s
OK	test_the_script_names_the_application_folder_it_works_in		NFR-013, TC-071	0.48 s

tests/system/test_health.py — 1 tests, 1 passed

US-001 acceptance criterion: app boots, health endpoint returns 200.

Result	Test	Purpose	Requirements	Time
OK	test_health_returns_ok		US-001	0.31 s

tests/system/test_i18n.py — 30 tests, 30 passed

I-22: the German interface. The catalogue must cover every string a template wraps in `()`,_

Result	Test	Purpose	Requirements	Time
OK	test_every_language_translates_the_whole_interface_inventory		FR-155, FR-167, TC-049	0.39 s
OK	test_every_template_string_has_a_german_entry		FR-155, FR-167, TC-049	0.49 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ar]		FR-155, FR-167, TC-049	0.96 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[bn]		FR-155, FR-167, TC-049	0.96 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[da]		FR-155, FR-167, TC-049	0.95 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[de]		FR-155, FR-167, TC-049	0.95 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[en]		FR-155, FR-167, TC-049	1.52 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[es]		FR-155, FR-167, TC-049	1.04 s

Result	Test	Purpose	Requirements	Time
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[fr]		FR-155, FR-167, TC-049	1.02 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[gu]		FR-155, FR-167, TC-049	1.02 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[hi]		FR-155, FR-167, TC-049	1.04 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[it]		FR-155, FR-167, TC-049	1.03 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ka]		FR-155, FR-167, TC-049	1.55 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[kn]		FR-155, FR-167, TC-049	1.04 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ml]		FR-155, FR-167, TC-049	1.04 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[mr]		FR-155, FR-167, TC-049	1.06 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[or]		FR-155, FR-167, TC-049	1.11 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[pa]		FR-155, FR-167, TC-049	1.04 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ta]		FR-155, FR-167, TC-049	0.94 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[te]		FR-155, FR-167, TC-049	1.48 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ur]		FR-155, FR-167, TC-049	0.96 s
OK	test_placeholders_survive_translation		FR-155, FR-167, TC-049	0.32 s
OK	test_the_browser_language_picks_the_login_page_language		FR-155, FR-167, TC-049	0.82 s
OK	test_the_catalogue_has_no_empty_or_identical_entries		FR-155, FR-167, TC-049	0.31 s
OK	test_the_choice_before_login_applies_to_the_login_page		FR-155, FR-167, TC-049	0.36 s
OK	test_the_german_login_page_keeps_english_credentials_hint_free_of_secrets		FR-155, FR-167, TC-049	0.36 s
OK	test_the_language_choice_is_remembered_for_the_user		FR-155, FR-167, TC-049	0.56 s
OK	test_the_user_preference_is_exposed_on_the_account_page		FR-155, FR-167, TC-049	0.47 s
OK	test_there_are_no_stale_german_entries_for_removed_template_strings		FR-155, FR-167, TC-049	0.56 s
OK	test_translate_falls_back_to_the_english_text		FR-155, FR-167, TC-049	0.32 s

tests/system/test_injection_and_session.py — 8 tests, 8 passed

TC-017, TC-026, TC-027, SEC-002, SEC-003, SEC-020-SEC-022: injection

Result	Test	Purpose	Requirements	Time
OK	test_audit_log_filters_resist_injection		SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027	0.42 s
OK	test_idle_session_expires_after_thirty_minutes	TC-017 / FR-002 / SEC-003.	FR-002, SEC-003, TC-017	0.42 s
OK	test_recent_activity_keeps_the_session_alive		SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027	0.41 s
OK	test_server_side_validation_ignores_client_side_constraints	SEC-020: every rule is enforced server-side, whatever the browser did.	SEC-020	0.69 s
OK	test_session_cookie_attributes	SEC-002 (the Secure flag is off only in the dev/test configs).	SEC-002	0.42 s
OK	test_session_expires_at_the_absolute_lifetime	SEC-003: 12 hours regardless of activity.	SEC-003	0.98 s
OK	test_sql_injection_probes_have_no_effect	TC-027 / SEC-021.	SEC-021, TC-027	0.95 s
OK	test_user_supplied_text_is_escaped_everywhere_it_renders	TC-026 / SEC-022.	SEC-022, TC-026	1.38 s

tests/system/test_logging_and_passwords.py — 8 tests, 8 passed

Security policy details from 11-security-architecture.md: §4 password policy

Result	Test	Purpose	Requirements	Time
OK	test_administrator_reset_also_honours_history			0.74 s
OK	test_authenticated_requests_record_the_user_id			0.41 s
OK	test_blocklist_ignores_comments_and_short_entries			0.00 s
OK	test_last_five_passwords_cannot_be_reused			2.90 s
OK	test_only_hashes_are_stored_in_the_history			0.49 s
OK	test_policy_length_and_blocklist			0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_requests_are_logged_as_json_without_credentials			0.41 s
OK	test_unhandled_errors_are_logged_with_the_request_id			0.46 s

tests/system/test_multiprocess.py — 4 tests, 4 passed

Two real server processes on one database file (docs/14-deployment-architecture.md §3):

Result	Test	Purpose	Requirements	Time
OK	test_concurrent_writers_in_two_processes_all_succeed		NFR-009, TC-050	15.47 s
OK	test_the_ip_limit_is_shared_and_other_users_are_unaffected_until_it_is_reached		NFR-009, TC-050	15.06 s
OK	test_the_limit_survives_a_restart		NFR-009, TC-050	15.01 s
OK	test_the_username_limit_is_shared_by_both_processes		NFR-009, TC-050	16.66 s

tests/system/test_no_cloud.py — 3 tests, 3 passed

US-120 / TC-029: block cloud-SDK imports, cloud identity providers and

Result	Test	Purpose	Requirements	Time
OK	test_no_cloud_sdk_imports_in_source		TC-029, US-120	0.85 s
OK	test_no_cloud_sdk_in_requirements		TC-029, US-120	0.01 s
OK	test_templates_and_static_reference_no_external_assets		TC-029, US-120	0.17 s

tests/system/test_openapi_conformance.py — 5 tests, 5 passed

EPIC-11 / FR-080: the code and docs/artifacts/openapi.yaml describe the

Result	Test	Purpose	Requirements	Time
OK	test_every_documented_operation_is_implemented_and_vice_versa		FR-080	0.33 s
OK	test_live_payloads_only_use_declared_properties		FR-080	0.89 s
OK	test_lookup_and_domain_request_payloads_match_contract		FR-080	0.78 s
OK	test_revision_diff_payload_matches_contract		FR-080	0.90 s
OK	test_unauthenticated_requests_are_rejected_for_every_documented_operation		FR-080	0.37 s

tests/system/test_permission_grid.py — 14 tests, 14 passed

TC-018 / TC-018a / TC-018b / TC-019 / FR-081 / SEC-010: the full

Result	Test	Purpose	Requirements	Time
OK	test_api_endpoints_require_authentication		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	0.82 s
OK	test_api_permission_grid[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.19 s
OK	test_api_permission_grid[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.14 s
OK	test_api_permission_grid[ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.53 s
OK	test_api_permission_grid[REVIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.11 s
OK	test_api_permission_grid[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.55 s
OK	test_api_permission_grid[VIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.03 s
OK	test_ui_permission_grid[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.76 s
OK	test_ui_permission_grid[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.16 s
OK	test_ui_permission_grid[ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.23 s
OK	test_ui_permission_grid[REVIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.14 s
OK	test_ui_permission_grid[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.65 s
OK	test_ui_permission_grid[VIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	0.91 s
OK	test_ui_redirects_anonymous_users_to_login		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	0.80 s

tests/system/test_production_config.py — 7 tests, 7 passed

SEC-002 / deployment hardening: a production instance must not start with a

Result	Test	Purpose	Requirements	Time
OK	test_development_and_testing_still_start_with_the_default_key		SEC-002	0.18 s
OK	test_production_refuses_a_weak_secret_key[]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[change-me-in-every-real-environment]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[dev-secret-key-change-me]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[too-short]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx]		SEC-002	0.00 s
OK	test_production_starts_with_a_strong_secret_key		SEC-002	0.10 s

tests/system/test_route_inventory.py — 13 tests, 13 passed

Every route is access-controlled, and the declared role lists are what is enforced.

Result	Test	Purpose	Requirements	Time
OK	test_a_tokenised_endpoint_is_always_a_get	API tokens may only ever read (docs/11 §12): allow_token must never sit on a write.	FR-081, SEC-010, TC-018, TC-048	0.33 s
OK	test_every_gated_route_refuses_anonymous_callers		FR-081, SEC-010, TC-018, TC-048	0.47 s
OK	test_every_route_declares_its_access_control		FR-081, SEC-010, TC-018, TC-048	0.30 s
OK	test_login_only_routes_are_reachable_by_every_role		FR-081, SEC-010, TC-018, TC-048	10.95 s
OK	test_login_page_credentials_still_work_for_the_ui_walk		FR-081, SEC-010, TC-018, TC-048	0.43 s
OK	test_role_lists_are_never_empty_and_only_name_real_roles		FR-081, SEC-010, TC-018, TC-048	0.30 s
OK	test_the_declared_role_list_is_what_each_role_gets[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-048	2.45 s
OK	test_the_declared_role_list_is_what_each_role_gets[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-048	1.62 s
OK	test_the_declared_role_list_is_what_each_role_gets[ENGINEER]		FR-081, SEC-010, TC-018, TC-048	1.65 s
OK	test_the_declared_role_list_is_what_each_role_gets[REVIEWER]		FR-081, SEC-010, TC-018, TC-048	2.19 s
OK	test_the_declared_role_list_is_what_each_role_gets[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-048	1.39 s
OK	test_the_declared_role_list_is_what_each_role_gets[VIEWER]		FR-081, SEC-010, TC-018, TC-048	1.21 s
OK	test_the_public_list_is_exactly_the_routes_that_need_no_session		FR-081, SEC-010, TC-018, TC-048	0.30 s

tests/system/test_security_headers.py — 4 tests, 4 passed

SEC-025: standard security headers on every response.

Result	Test	Purpose	Requirements	Time
OK	test_api_response_has_security_headers		SEC-025	0.30 s
OK	test_hsts_sent_over_https		SEC-025	0.31 s
OK	test_templates_have_no_inline_script_or_handlers	The CSP forbids inline script; guard against regressions.	SEC-025	0.06 s
OK	test_ui_page_has_csp_and_no_hsts_over_plain_http		SEC-025	0.38 s

tests/system/test_sqlite_path.py — 2 tests, 2 passed

A new machine: the database file is found, and its folder created, from the example .env

Result	Test	Purpose	Requirements	Time
OK	test_a_relative_sqlite_path_is_read_from_the_application_directory		NFR-013	0.13 s
OK	test_an_absolute_path_gets_its_folder_and_memory_is_left_alone		NFR-013	0.21 s

tests/system/test_translations_doc.py — 27 tests, 27 passed

The translation catalogue document and the review sheets for native speakers

Result	Test	Purpose	Requirements	Time
OK	test_every_language_has_its_section_and_its_coverage_row		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.12 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[ar-right]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.92 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[fr-left]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.49 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[ur-right]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.50 s
OK	test_string_ids_are_stable_and_never_collide		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.06 s
OK	test_the_command_line_writes_one_sheet_per_language_and_refuses_unknown_ones		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	8.06 s
OK	test_the_concept_names_the_layers_the_checks_and_the_open_items		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.01 s
OK	test_the_document_is_exactly_what_the_generator_produces_from_the_code		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.27 s
OK	test_the_fingerprint_changes_when_one_translation_changes		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.01 s
OK	test_the_review_sheet_of_every_language_is_complete[ar]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_review_sheet_of_every_language_is_complete[bn]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.58 s
OK	test_the_review_sheet_of_every_language_is_complete[da]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.55 s
OK	test_the_review_sheet_of_every_language_is_complete[de]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.07 s
OK	test_the_review_sheet_of_every_language_is_complete[es]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.57 s
OK	test_the_review_sheet_of_every_language_is_complete[fr]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.55 s
OK	test_the_review_sheet_of_every_language_is_complete[gu]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_review_sheet_of_every_language_is_complete[hi]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_review_sheet_of_every_language_is_complete[it]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.57 s
OK	test_the_review_sheet_of_every_language_is_complete[ka]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_review_sheet_of_every_language_is_complete[kn]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_review_sheet_of_every_language_is_complete[ml]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.62 s
OK	test_the_review_sheet_of_every_language_is_complete[mr]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.03 s
OK	test_the_review_sheet_of_every_language_is_complete[or]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.57 s
OK	test_the_review_sheet_of_every_language_is_complete[pa]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.05 s
OK	test_the_review_sheet_of_every_language_is_complete[ta]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.58 s
OK	test_the_review_sheet_of_every_language_is_complete[te]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.57 s
OK	test_the_review_sheet_of_every_language_is_complete[ur]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.96 s

tests/system/test_uat.py — 14 tests, 14 passed

TC-030: UAT-01 .. UAT-13 (docs/13-test-strategy.md §7) as scripted

Result	Test	Purpose	Requirements	Time
OK	test_uat_01_new_code_has_no_identifier_until_submitted		UAT-01	1.21 s
OK	test_uat_02_either_signoff_order_reaches_pending_approval_automatically		UAT-02	1.58 s
OK	test_uat_03_chief_engineer_approval_visible_to_the_engineer		UAT-03	1.49 s
OK	test_uat_04_rejection_comment_visible_and_prior_signoff_cleared		UAT-04	1.33 s
OK	test_uat_05_cross_domain_signoff_unblocks_submit		UAT-05	1.00 s
OK	test_uat_06_release_build_excludes_unselected_and_sandbox		UAT-06	1.99 s
OK	test_uat_07_viewer_searches_and_exports_released_catalogue_for_one_group		UAT-07	1.36 s
OK	test_uat_08_sandbox_lifecycle_and_traceable_deletion		UAT-08	0.72 s
OK	test_uat_09_nobody_can_decide_their_own_submission		UAT-09	1.59 s
OK	test_uat_09b_chief_engineer_cannot_be_a_prior_signer_or_author		TC-030, UAT-01, UAT-13	1.28 s
OK	test_uat_10_full_database_export_is_administrator_only		UAT-10	0.97 s
OK	test_uat_11_import_creates_only_governed_drafts_and_an_error_report		UAT-11	0.77 s
OK	test_uat_12_subgroup_is_mandatory_and_overlaps_are_refused		UAT-12	0.75 s
OK	test_uat_13_failed_form_keeps_input_and_dropdowns_explain_themselves		UAT-13	0.88 s

tests/system/test_version.py — 4 tests, 4 passed

The version of the application is written once and shown where a person or a monitor looks for

Result	Test	Purpose	Requirements	Time
OK	test_the_api_contract_documents_the_version_in_both_health_answers		FR-184, NFR-013, TC-049	0.02 s
OK	test_the_health_endpoints_name_the_version		FR-184, NFR-013, TC-049	0.40 s
OK	test_the_login_page_and_the_menu_show_the_version		FR-184, NFR-013, TC-049	0.94 s
OK	test_the_version_is_a_release_number_and_pyproject_agrees		FR-184, NFR-013, TC-049	0.02 s

tests/system/test_windows_gateway.py — 7 tests, 7 passed

The Windows sign-in against a real Windows identity (docs/03-srs.md FR-163, docs/11 SEC-033).

Result	Test	Purpose	Requirements	Time
OK	test_a_header_the_client_sends_is_dropped_by_the_gateway		FR-163, SEC-033, TC-062	0.92 s
OK	test_a_request_without_credentials_gets_a_challenge_and_never_reaches_the_application		FR-163, SEC-033, TC-062	1.97 s
OK	test_an_unknown_windows_user_and_a_deactivated_account_are_refused		FR-163, SEC-033, TC-062	0.95 s
OK	test_sspi_names_the_windows_user_who_runs_the_process[NTLM]		FR-163, SEC-033, TC-062	0.00 s
OK	test_sspi_names_the_windows_user_who_runs_the_process[Negotiate]		FR-163, SEC-033, TC-062	0.00 s
OK	test_the_header_is_ignored_when_it_does_not_come_from_the_trusted_gateway		FR-163, SEC-033, TC-062	1.39 s
OK	test_the_whole_chain_signs_the_windows_user_in		FR-163, SEC-033, TC-062	0.92 s

tests/performance/test_performance.py — 9 tests, 9 passed

NFR-001, TC-024, TC-025, NFR-004: performance targets on a temp-file SQLite

Result	Test	Purpose	Requirements	Time
OK	test_concurrent_identifier_allocation_in_one_group	TC-025: parallel submissions in one popular group yield unique, gap-free	TC-025	0.84 s
OK	test_export_issues_a_constant_number_of_queries	Regression guard for the N+1 problem: exporting 500 revisions must not cost	NFR-001, NFR-004, TC-024, TC-025	0.77 s
OK	test_export_of_5000_records_within_target[csv-5]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	1.84 s

Result	Test	Purpose	Requirements	Time
OK	test_export_of_5000_records_within_target[json-5]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	1.84 s
OK	test_export_of_5000_records_within_target[pdf-15]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	7.08 s
OK	test_export_of_5000_records_within_target[xlsx-15]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	3.15 s
OK	test_fifty_concurrent_users_are_served_without_errors	NFR-004: 50 concurrent named users (each searching every ~8 s —	NFR-004	28.39 s
OK	test_reads_are_not_blocked_by_an_open_write_transaction	WAL read concurrency: a reader proceeds while another connection holds	NFR-001, NFR-004, TC-024, TC-025	0.62 s
OK	test_search_and_read_latency_at_20000_revisions	TC-024.	TC-024	6.54 s

9. Run history

#	Date (UTC)	Commit	Order	Tests	Passed	Failed	Retried	Coverage	Gates	Duration
1	2026-10-21 16:36	f23ef18*	reverse	318	318	0	1	88%	5/5 pass	232 s
2	2026-10-21 16:42	f23ef18*	shuffle:11	318	318	0	0	88%	5/5 pass	235 s
3	2026-10-21 16:49	f23ef18*	file	318	318	0	1	88%	5/5 pass	237 s
4	2026-10-21 16:57	f23ef18*	file	318	318	0	0	88%	5/5 pass	244 s
5	2026-10-21 17:09	f23ef18*	shuffle:23	319	319	0	0	88%	5/5 pass	242 s
6	2026-10-21 17:15	f23ef18*	file	319	319	0	0	88%	5/5 pass	237 s
7	2026-10-22 10:02	f23ef18*	file	329	329	0	0	89%	5/5 pass	150 s
8	2026-10-23 06:16	c0aff94*	file	601	600	1	0	92%	5/5 pass	589 s
9	2026-10-23 06:27	c0aff94*	reverse	601	600	1	0	92%	5/5 pass	526 s
10	2026-10-23 06:38	c0aff94*	shuffle:7	601	600	1	0	92%	5/5 pass	473 s
11	2026-10-23 06:50	c0aff94*	file	602	602	0	0	92%	5/5 pass	472 s
12	2026-10-23 06:59	c0aff94*	reverse	602	602	0	0	92%	5/5 pass	470 s
13	2026-10-23 07:08	c0aff94*	shuffle:7	602	602	0	0	92%	5/5 pass	469 s
14	2026-10-23 07:18	c0aff94*	file+filedb	602	601	1	0	92%	5/5 pass	542 s
15	2026-10-23 07:31	c0aff94*	file+filedb	602	602	0	0	92%	5/5 pass	590 s
16	2026-10-23 07:41	c0aff94*	file	602	602	0	0	92%	5/5 pass	505 s
17	2026-10-23 07:54	c0aff94*	reverse	607	607	0	0	92%	5/5 pass	459 s
18	2026-10-23 08:04	c0aff94*	shuffle:7	607	607	0	0	92%	5/5 pass	491 s
19	2026-10-23 08:16	c0aff94*	file+filedb	607	607	0	0	92%	5/5 pass	628 s
20	2026-10-23 08:25	c0aff94*	file	607	607	0	0	92%	5/5 pass	508 s
21	2026-10-23 15:33	c0aff94*	file	807	807	0	0	92%	5/5 pass	685 s
22	2026-10-23 19:53	3670346*	file	830	828	2	0	93%	2/5 pass	850 s
23	2026-10-23 20:10	3670346*	file	830	830	0	0	93%	5/5 pass	728 s
24	2026-10-23 23:30	9f038bc*	file	848	847	1	0	93%	5/5 pass	741 s
25	2026-10-23 23:47	9f038bc*	file	848	848	0	0	93%	5/5 pass	748 s
26	2026-10-24 10:42	e07a82d*	file	867	867	0	0	93%	4/5 pass	768 s
27	2026-10-24 10:57	88bf318*	file	867	867	0	0	93%	5/5 pass	772 s
28	2026-10-24 11:50	9828815*	file	876	876	0	0	93%	5/5 pass	773 s
29	2026-10-24 12:42	1f1f0c9*	file	892	891	1	0	93%	5/5 pass	807 s
30	2026-10-24 12:58	1f1f0c9*	file	892	892	0	0	93%	5/5 pass	806 s
31	2026-10-24 14:37	a81555a*	file	904	904	0	0	93%	4/5 pass	967 s
32	2026-10-24 14:56	a81555a*	file	915	915	0	0	93%	5/5 pass	979 s
33	2026-10-24 16:00	2cd8fe6*	file	927	927	0	0	93%	5/5 pass	993 s
34	2026-10-24 17:31	afc84a3*	file	946	945	1	0	93%	3/5 pass	1051 s
35	2026-10-24 17:53	afc84a3*	file	946	943	3	0	93%	5/5 pass	1050 s
36	2026-10-24 18:12	afc84a3*	file	946	946	0	0	93%	5/5 pass	1054 s
37	2026-10-25 11:41	016e74f*	file	956	956	0	0	93%	3/5 pass	1207 s
38	2026-10-25 12:06	016e74f*	file	956	956	0	0	93%	5/5 pass	1205 s

#	Date (UTC)	Commit	Order	Tests	Passed	Failed	Retried	Coverage	Gates	Duration
39	2026-10-26 09:59	b0029bd*	file	958	958	0	0	93%	5/5 pass	1163 s
40	2026-10-26 10:26	2a5a8ff	file	966	966	0	0	93%	5/5 pass	1113 s
41	2026-10-26 11:39	97af72c	file	973	973	0	0	93%	5/5 pass	1099 s
42	2026-10-26 12:29	a62d0c6*	file	1007	1007	0	0	93%	5/5 pass	1209 s